Dual Sparse Representation Graph-Based Copropagation for Semisupervised Hyperspectral Image Classification

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***Abstract*— Graph-based semisupervised hyperspectral image (HSI) classification methods have obtained extensive attention. In graph-based methods, a graph is first constructed, and then the label propagation is carried out on the constructed graph to obtain the labels for unknown samples. However, the results of label propagation may be unreliable, especially in the case of very limited labeled samples. To address the above problem, we propose dual sparse representation graph-based collaborative propagation (DSRG-CP) for HSI classification. Specifically, DSRG-CP adopts sparse representation (SR) to construct spectral and spatial graphs on spectral and spatial dimensions, respectively. Then, label propagation is performed on two graphs iteratively. In each iteration, only the samples with high classification confidence from one graph are added into another graph as labeled samples for the next label propagation. After several iterations, the labels of unlabeled samples are predicted by fusing the results of label propagation from two graphs. In addition, to make the spectral graph more discriminative, the regularizer of spectral statistical information is added into spectral SR model. To make the classification results more consistent in space, the superpixel block constraint is added into spatial graph model as regularizer. To evaluate the performance of the proposed method, DSRG-CP is compared with several graph-based methods and other state-of-the-art methods. Extensive experiments on real HSI data sets show that DSRG-CP can obtain competitive results for HSI classification.**

***Index Terms*— Collaborative propagation (copropagation), hyperspectral image (HSI) classification, semisupervised learning (SSL), sparse representation (SR) graph.**

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I. Introduction

CLASSIFICATION is an essential task of hyperspectral image (HSI) analysis. In the past few years, many supervised methods, such as k-nearest neighbors (kNNs) [1], [2], support vector machines (SVMs) [3], [4], sparse representation-based classifiers (SRCs) [2], [5], extreme learning machines (ELMs) [6], [7], and random forests (RFs) [8], [9], were widely used for HSI classification. These methods are faced with the imbalance between limited labeled samples and high dimensionality of hyperspectral data, which easily result in Hughes phenomenon [10]. Semisupervised learning (SSL) methods [11] provide promising solutions to address the above problem. Several SSL methods have been successfully used for HSI classification, which are categorized into five classes: 1) self-training [12], [13]; 2) generative models [14]; 3) transductive SVM (TSVM) [15]; 4) graph-based methods [16], [17]; and 5) cotraining [18], [19]. Our work concentrates on graph-based approaches, which have achieved great success in HSI classification [16], [17], [20]–[29].

There are two important parts of graph-based SSL. One is the construction of the graph, where the vertices in the graph represent data points (both unlabeled and labeled), and the edge weights indicate the similarities between pairwise samples. The other is propagating the labels from labeled samples to unlabeled samples via the graph. Traditional graph-based SSL methods generally use kNN or ε-ball neighborhood to obtain the adjacency matrix of the graph, then several different ways such as Gaussian kernel (GK) function [16], [21], [30], [31], local manifold learning (LML) [22], [24], and local linear reconstruction (LLR) [32], [33] are adopted to obtain the edge weights.

Recently, sparse representation (SR) was introduced to construct graph-based semisupervised HSI classification models [23], [26], [28]. Cheng *et al.* [34] and Yan and Wang [35] first used SR to construct l_1-graphs, where the similarities among samples are measured by solving l_1 optimization. Then, SR graph-based SSL methods were used for HSI classification. For example, Gu and Feng [36] firstly used l_1-graph for semisupervised HSI classification. Shao *et al.* [23] proposed to utilize a probabilistic class structure to regularize the SR graph. Shao *et al.* [26] also adopted spatial information to regularize the SR graph. In addition, SR graphs were used for discriminant analysis of HSIs [37], [38]. SR graph has a solid mathematical background, and it can determine the graph

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adjacency relationships and compute the similarities among data points, simultaneously. In this article, we also adopt SR to construct graphs.

In graph-based SSL approaches, the labels of unlabeled samples are obtained by directly propagating the labels from labeled samples to unlabeled samples via the graph. However, previous methods only perform label propagation once, which leads to unreliable results when very limited samples are labeled. Our motivation is to perform label propagation iteratively to enlarge the number of labeled samples, and then to use the expanded labeled samples for label propagation. In SSL paradigm, self-training [12], [13], [39] and cotraining [18], [19], [40]–[43] can obtain the pseudolabels of unlabeled samples iteratively. Considering that HSIs can be characterized by spectral view and spatial view independently, we use the cotraining style for the iteration of label propagation. We refer to this strategy as collaborative propagation (copropagation).

Copropagation performs label propagation on two different views alternatively, which needs to satisfy the necessary condition of view independence, whereas spectral view and spatial view happen to be independent of each other. So, we use spectral and spatial features to construct spectral and spatial SR graphs, respectively, then the labels are propagated on two graphs by means of collaboration. Different from cotraining, which need to train models iteratively, the copropagation only performs label propagation on two graphs alternately.

To make the constructed graph better represent the relationship between samples, several methods incorporated with different regularizations were proposed for HSI classification. For example, Ma *et al.* [24] adopted spatial information to regularize the objective function of graph model. Shao *et al.* [26] utilized spatial information and class structure to regularize the SR graph. Jia *et al.* [28] introduced $l_{1/2}$ norm regularization to describe the low-rank structure of the decomposed data more precisely. Although these methods added regularizations into graph models, they ignore the rich spectral statistical information and image neighborhood structure information. Whereas spectral statistical information can increase discrimination [44], the image neighborhood structure information can make the classification results more consistent [27]. In this work, we employ the spectral and spatial information fully to increase the discrimination of SR graph and make the classification results more consistent. Specifically, the spectral statistical similarity constraint and superpixel block similarity constraint are added into spectral SR graph model and spatial SR graph model, respectively.

Inspired by the above analysis, in this article, we propose dual sparse representation graph-based collaborative propagation (DSRG-CP) for semisupervised HSI classification. First, we construct two individual SR graphs. Specifically, when constructing the spectral graph (SPE-G) by using spectral features, the regularizer of spectral statistical information is added into the objective function, which increases the discriminant ability of the model. When constructing spatial graph (SPA-G) by using spatial features, the regularizer of the superpixel block structure is added into the objective function, which makes the classification results more consistent in the spatial neighborhood. The objective functions of both SPE-G and SPA-G are solved by alternating direction methods with an adaptive penalty (ADMAP) [45]. Second, we perform copropagation on two graphs iteratively. At each iteration, the samples with high classification confidence are added into another graph as labeled samples for the next label propagation. At last, after several iterations, the propagation will become stable, then the results of label propagation from two graphs are fused to predict the class labels. The main contributions of this article are threefolds.

1) The DSRG-CP is proposed for semisupervised HSI classification.
2) To increase the discrimination of spectral SR graph, the spectral statistical information constraint is added into SR model as regularizer.
3) The superpixel block constraint is added into spatial SR graph model as regularizer, which makes the graph representation more consistent in spatial neighborhood.

The rest of this article is organized as follows. Section II introduces related work. The proposed method is detailed in Section III. Section IV exhibits the experimental results. At last, the conclusions of our work are shown in Section V.

II. Related Work

In this section, we first review graph-based SSL framework including the construction of the graph and label propagation. Then, the main idea of SR graph is introduced.

A. Graph-Based SSL Framework

Traditional graph construction includes the determination of graph adjacency relationships and the computation of edge weights. Generally, two well-known methods including kNN and $.\varepsilon$-ball neighborhood are used to determine adjacency relationships [11], [30].

Once the adjacency relationships are determined, several approaches can be used to compute the edge weights of the graph. These methods include GK function [16], [21], [30], [31], LML [22], [24] and nonnegative LLR [32], [33].

After constructing the graph, one can obtain the labels of unlabeled samples through label propagation methods. Popular label propagation approaches include Gaussian fields and harmonic functions (GFHFs) [30], local and global consistency (LGC) [31], and Tikhonov regularization [46]. In this article, we use GFHF for label propagation in our method. Next, we introduce the principle of GFHF.

Let $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_l, \mathbf{x}_{l+1}, \cdots \mathbf{x}_{l+u}\}$ be the data, where $\mathbf{X}_l = \{\mathbf{x}_1, \ldots, \mathbf{x}_l\}$ are labeled samples, $\mathbf{X}_u = \{\mathbf{x}_{l+1}, \cdots \mathbf{x}_{l+u}\}$ are unlabeled samples, and $l + u = n$. The class label of each labeled sample belongs to $\mathbf{\Omega} = \{\Omega_1, \ldots, \Omega_c\}$. Let $\mathbf{Y} \in \mathbb{R}_*^{n\times c}$ be a class probability matrix, where $\mathbb{R}_*$ represents nonnegative real value space. Initially, we set $\mathbf{Y} = \mathbf{0}$. Then, for each labeled sample $\mathbf{x}_i \in \mathbf{X}_l$, we set $\mathbf{Y}_{ij} = 1$ if $\mathbf{x}_i$ belongs to class $\mathbf{\Omega}_j$. GFHF aims to learn the function $\mathbf{F} = [\mathbf{F}_l; \mathbf{F}_u] \in \mathbb{R}_*^{n\times c}$ by minimizing the following function, while constraining the predicted label information of labeled samples to be equal to

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the real label information, i.e., $\mathbf{F}_l = \mathbf{Y}_l$:

$$\min_{\mathbf{F}} \frac{1}{2}\sum_{i,j=1}^{n} \mathbf{W}_{ij}\left\|\mathbf{f}_i - \mathbf{f}_j\right\|^2 \quad \text{s.t. } \mathbf{F}_l = \mathbf{Y}_l \tag{1}$$

where $\mathbf{W}$ is the weight matrix of the graph, $(1/2)\sum_{i,j=1}^{n} \mathbf{W}_{ij}\left\|\mathbf{f}_i - \mathbf{f}_j\right\|^2 = \mathrm{Tr}(\mathbf{F}^{\mathrm{T}}\mathbf{L}\mathbf{F})$ is called graph Laplacian regularization, and $\mathbf{f}_i \in \mathbb{R}_*^{1\times c}$ and $\mathbf{f}_j \in \mathbb{R}_*^{1\times c}$ are the prediction probability vectors of samples $\mathbf{x}_i$ and $\mathbf{x}_j$, respectively. $\mathbf{L} = \mathbf{D} - \mathbf{W}$ is the Laplacian matrix, where $\mathbf{D}$ is a diagonal degree matrix given by $\mathbf{D}_{ii} = \sum_j \mathbf{W}_{ij}$.

We can divide the matrix $\mathbf{L}$ into four submatrices according to the labeled and unlabeled vertices. Then, (1) becomes

$$\min_{\mathbf{F}_u}[\mathbf{Y}_l\mathbf{F}_u]\begin{bmatrix}\mathbf{L}_{ll} & \mathbf{L}_{lu}\\ \mathbf{L}_{ul} & \mathbf{L}_{uu}\end{bmatrix}\begin{bmatrix}\mathbf{Y}_l^{\mathrm{T}}\\ \mathbf{F}_u^{\mathrm{T}}\end{bmatrix} \quad \text{s.t. } \mathbf{F}_l = \mathbf{Y}_l. \tag{2}$$

By setting the partial derivatives to zero with respect to $\mathbf{F}_u$, we can obtain the closed form solution of (2)

$$\mathbf{F}_u = -\mathbf{L}_{uu}^{-1}\mathbf{L}_{ul}\mathbf{Y}_l. \tag{3}$$

B. SR Graph

SR has strong mathematical theory, which has shown significant performance on graph-based semisupervised HSI classification. Cheng *et al.* [34] and Yan and Wang [35] first proposed SR graph models. Given n samples $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \cdots \mathbf{x}_n\} \in \mathbb{R}^{d\times n}\}$, for each $\mathbf{x}_i \in \mathbf{X}$, SR graphs seeks to encode it as the linear combination of other samples in $\mathbf{X}$. Generally, it is formulated as the following ℓ_1 optimization problem [34], [35]:

$$\begin{aligned}\mathbf{W} = \ & \arg\min\|\mathbf{W}\|_1\\ & \text{s.t. } \mathbf{X} = \mathbf{X}\mathbf{W}, \mathrm{diag}(\mathbf{W}) = 0, \quad \mathbf{W} \geq 0\end{aligned} \tag{4}$$

where $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \cdots \mathbf{w}_n]$ is a coefficient matrix, each $\mathbf{w}_i$ indicates the representation coefficient of $\mathbf{x}_i$. The constraint $\mathrm{diag}(\mathbf{W}) = 0$ avoids the case of self-similarity. The coefficient $\mathbf{w}_{ij}$ reflects the similarity between $\mathbf{x}_i$ and $\mathbf{x}_j$.

To make the model robust on the noisy data, the above optimization can be extended as follows [47]:

$$\begin{aligned}\mathbf{W} = \ & \arg\min\frac{1}{2}\|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda\|\mathbf{W}\|_1\\ & \text{s.t. } \mathrm{diag}(\mathbf{W}) = 0, \quad \mathbf{W} \geq 0\end{aligned} \tag{5}$$

where parameter λ is used to balance the sparsity and the reconstruction error.

Several SR graphs were proposed for semisupervised HSI classification [23], [26], [32] and discriminant analysis [37], [38]. In this article, we use SR to construct SPE-G and SPA-G on spectral and spatial views, respectively. To make the SPE-G more discriminative, we add spectral statistical information as regularizer into SPE-G model. To make the classification results more consistent in spatial neighborhood, we add superpixel block structure as regularizer into SPA-G model. At last, we perform copropagation on two graphs to obtain the final classification results.

III. DSRG-CP

A. Spectral Graph

Previous spectral graph models only use spectral features to measure the similarities between pairwise samples, which do not consider the spectral statistical information. However, spectral statistical information plays an essential role in HSI analysis, which can increase the discriminant ability [44]. Spectral information divergence (SID) [44], relying on information theory, is a typical type of spectral statistical information, and it has been proved very useful on spectral discrimination [48]. In our spectral graph, we add SID as regularizer into the SR model.

The SID between samples $\mathbf{x}_i = (x_{i1}, x_{i2}, \cdots x_{id})^{\mathrm{T}}$ and $\mathbf{x}_j = \left(x_{j1}, x_{j2}, \cdots x_{jd}\right)^{\mathrm{T}}$ is defined as follows:

$$\mathbf{SID}\left(\mathbf{x}_i, \mathbf{x}_j\right) = \sum_{k=1}^{d} r_k \log(r_k/s_k) + \sum_{k=1}^{d} s_k \log(s_k/r_k) \tag{6}$$

where $r_k = x_{ik}/\sum_{t=1}^{d} x_{it}$, $s_k = x_{jk}/\sum_{t=1}^{d} x_{jt}$, and d represents the spectral dimension.

To make the SPE-G more discriminative, we add the SID into the spectral SR model. We define the regularizer as

$$S(\mathbf{W}) = \sum_{ij}\left|\mathbf{W}_{ij}\right|\left|\mathbf{SID}_{ij}\right| = \|\mathbf{SID} \otimes \mathbf{W}\|_1 \tag{7}$$

where $\otimes$ represents bitwise multiplication. The $\mathbf{SID}_{ij}$ represents the statistical similarity between sample $\mathbf{x}_i$ and sample $\mathbf{x}_j$, and a large value of $\mathbf{SID}_{ij}$ indicates that $\mathbf{x}_i$ and $\mathbf{x}_j$ are more statistical dissimilar. When minimizing (7), it prefers to assign smaller weights to the samples with large statistical dissimilar.

We add (7) into (5), then the following SPE-G model is formulated:

$$\begin{aligned}&\min_{\mathbf{W}} \frac{1}{2}\|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda_1\|\mathbf{W}\|_1 + \lambda_2 S(\mathbf{W})\\ &\text{s.t. } \mathrm{diag}(\mathbf{W}) = 0, \quad \mathbf{W} \geq 0\end{aligned} \tag{8}$$

where $\lambda_1 > 0$ is used to balance the sparsity and the reconstruction error, and $\lambda_2 > 0$ is a parameter to control the weight of regularizer. We use the ADMAP [45] to solve (8). According to [45], we introduce auxiliary variable $\mathbf{P}$ to separate (8), then the objective function is formulated as

$$\begin{aligned}&\min_{\mathbf{W},\mathbf{J}} \frac{1}{2}\|\mathbf{X} - \mathbf{X}\mathbf{P}\|_F^2 + \lambda_1\|\mathbf{W}\|_1 + \lambda_2\|\mathbf{SID} \otimes \mathbf{W}\|_1\\ &\text{s.t. } \mathbf{P} = \mathbf{W} - \mathrm{diag}(\mathbf{W}), \quad \mathbf{W} \geq 0.\end{aligned} \tag{9}$$

Then, the augmented Lagrangian function of (9) is formulated as

$$\begin{aligned}L(\mathbf{W}, \mathbf{P}, \mathbf{\Lambda}, \mu) = & \frac{1}{2}\|\mathbf{X} - \mathbf{X}\mathbf{P}\|_F^2 + \lambda_1\|\mathbf{W}\|_1\\ & + \lambda_2\|\mathbf{SID} \otimes \mathbf{W}\|_1 + \frac{\mu}{2}\|\mathbf{P} - \mathbf{W} + \mathrm{diag}(\mathbf{W})\|_F^2\\ & + \mathrm{tr}(\mathbf{\Lambda}^{\mathrm{T}}(\mathbf{P} - \mathbf{W} + \mathrm{diag}(\mathbf{W})))\end{aligned} \tag{10}$$

where μ is a numerical parameter, and $\mathbf{\Lambda}$ is the Lagrange multiplier.

To minimize (10), we update variables $\mathbf{W}$, $\mathbf{P}$, and $\mathbf{\Lambda}$ alternately. Detailed updating steps are as follows.

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1) Fix other variables and update $\mathbf{P}$

$$\mathbf{P} = \arg\min_{\mathbf{P}} \frac{1}{2}\|\mathbf{X} - \mathbf{X}\mathbf{P}\|_F^2 + \frac{\mu}{2}\|\mathbf{P} - \mathbf{W}\|_F^2 + \mathrm{tr}(\mathbf{\Lambda}^{\mathrm{T}}(\mathbf{P} - \mathbf{W})). \quad (11)$$

We set the partial derivative of (11) with respect to $\mathbf{P}$ to zero, and we obtain the solution of (11)

$$\mathbf{P} = \left(\mathbf{X}^{\mathrm{T}}\mathbf{X} + \mu\mathbf{I}\right)^{-1}(\mathbf{X}^{\mathrm{T}}\mathbf{X} - \mu\mathbf{W} - \mathbf{\Lambda}). \quad (12)$$

2) Fix other variables and update $\mathbf{W}$

$$\mathbf{W} = \arg\min_{\mathbf{W}} \left\|(\lambda_1\mathbf{1}^{\mathrm{T}} + \lambda_2\mathbf{SID}) \otimes \mathbf{W}\right\|_1 + \frac{\mu}{2}\left\|\mathbf{W} - \mathrm{diag}(\mathbf{W}) - \left(\mathbf{P} + \frac{\mathbf{\Lambda}}{\mu}\right)\right\|_F^2. \quad (13)$$

The solution of (13) is

$$\mathbf{W} = \tilde{\mathbf{W}} - \mathrm{diag}(\tilde{\mathbf{W}}) \quad (14)$$

where each entry of $\tilde{\mathbf{W}}$ is computed through the following equation:

$$\tilde{\mathbf{W}}_{ij} = \max\left(\Theta_{((\lambda_1+\lambda_2\mathbf{SID}_{ij})\mu^{-1})}\left(\left(\mathbf{P} + \frac{\mathbf{\Lambda}}{u}\right)_{ij}\right), 0\right) \quad (15)$$

where $\Theta_{(\varepsilon)}(t) = \max(|t| - \varepsilon, 0)\mathrm{sgn}(t)$ is a soft thresholding function and $\mathrm{sgn}(t)$ is sign function.

3) Fix other variables and update $\mathbf{\Lambda}$

$$\mathbf{\Lambda} = \mathbf{\Lambda} + \mu(\mathbf{P} - \mathbf{W}). \quad (16)$$

The procedure of solving (8) is described in Algorithm 1.

Algorithm 1 Solving Problem (8) by ADMAP

Input: data points $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \cdots \mathbf{x}_n\} \in \mathbb{R}^{d\times n}$, parameters $\lambda_1 > 0$, $\lambda_2 > 0$.

1: **Initialization:** $\mathbf{W} = \mathbf{0}$, $\mathbf{P} = \mathbf{0}$, $\mathbf{\Lambda} = \mathbf{0}$, $\mu^{(0)} = 0.1$, $\mu_{\max} = 10^{10}$, $\rho_0 = 1.1$, $\varepsilon_1 = 10^{-4}$, $\varepsilon_2 = 10^{-4}$, $\eta = \|\mathbf{X}\|_2^2 + 1$, $k = 0$.

2: Compute **SID** for all pairwise samples through (6). Feed $\mathbf{X}$ and **SID** into (8).

3: **While** $\|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F / \|\mathbf{X}\|_F \geq \varepsilon_1$ or $\mu_k \max(\sqrt{\eta}\|\mathbf{W}_k - \mathbf{W}_{k-1}\|_F, \|\mathbf{P}_k - \mathbf{P}_{k-1}\|_F)/\|\mathbf{X}\|_F \geq \varepsilon_2$ **do**

4: Update $\mathbf{P}_{k+1}$ by (12);

5: Update $\mathbf{W}_{k+1}$ by (14) and (15);

6: Update $\mathbf{\Lambda}_{k+1}$ by (16);

7: Update $\mu^{(k+1)}$: $\mu^{(k+1)} = \min(\mu_{\max}, \rho\mu^{(k)})$, where

$$\rho = \begin{cases} \rho_0, \text{if } \mu^{(k)}\max(\sqrt{\eta}\|\mathbf{W} - \mathbf{W}_{k-1}\|_F, \\ \quad \|\mathbf{P} - \mathbf{P}_{k-1}\|_F)/\|\mathbf{X}\|_F < \varepsilon_2 \\ 1, \text{otherwise.} \end{cases}$$

8: Update k: $k \leftarrow k + 1$;

9: **End while**

10: **Output:** The coefficient matrix $\mathbf{W}$.

Let $G{=}(V, E)$ be a connected graph, where vertexes V and edges E represent the samples and similarity relationships among samples, respectively. So, the key to creating a graph is to construct an adjacency matrix of the given data $\mathbf{X}$.

For the given data matrix $\mathbf{X}$, we use Algorithm 1 to obtain the coefficient matrix $\hat{\mathbf{W}}_1$ of $\mathbf{X}$, where each entry of the matrix $\hat{\mathbf{W}}_1$ represents the similarity between two samples. Since the matrix $\hat{\mathbf{W}}_1$ is asymmetric, we use (17) to obtain the symmetric matrix $\mathbf{W}^1$. Then, the matrix $\mathbf{W}^1$ is regarded as the adjacency matrix of SPE-G

$$\mathbf{W}^1 = (\hat{\mathbf{W}}_1 + \hat{\mathbf{W}}_1^{\mathrm{T}})\Big/2. \quad (17)$$

B. Spatial Graph

The spatial feature plays an important role in HSI analysis. Previous graph-based methods generally combine spectral and spatial features to construct spectral-spatial graphs or treat the spatial features as additional conditions to assist classification models. In this article, we utilize Gabor filter to extract the 2-D Gabor features from the first three principal components of original bands. Then, the extracted 2-D Gabor features are used to construct the spatial SR graph model. Gabor filter has been widely used to extract texture features and further used for HSI classification [21], [41]. For more information about the Gabor filter, please refer to [49].

Considering that the abundant structure information of HSIs can help the classification results more consistent in spatial neighborhood, we add the neighborhood structure information as constraint into SPA-G. Specifically, we use the superpixel method to obtain the image structure information. Superpixel method clusters spatially adjacent pixels with similar structures into an image block [50], and the superpixel method has been successfully used for HSI classification. Our goal is to let similar blocks have similar representation coefficients. Therefore, when we calculate the similarity of two pixels, we add the similarity of the corresponding blocks of the two pixels as a regularizer to the graph model.

We adopt the sum of minimum distance (SMD) [24], [51] to measure the similarity between two superpixel blocks. Suppose that $\mathbf{x}_i$ and $\mathbf{x}_j$ belong to superpixel $\mathbf{B}_i$ and superpixel $\mathbf{B}_j$ respectively, where $\mathbf{B}_i$ and $\mathbf{B}_j$ contain m and n pixels respectively. We first compute the distance between any two pixels in $\mathbf{B}_i$ and $\mathbf{B}_j$ and arrange these distances into a matrix $\mathbf{T} \in \mathbb{R}^{m\times n}$. Then, we compute SMD as follows:

$$d_{\mathrm{smd}}(\mathbf{B}_i, \mathbf{B}_j) = \frac{1}{m}\sum_{i=1}^{m}\min(\mathbf{T}_{i:}) + \frac{1}{n}\sum_{j=1}^{n}\min(\mathbf{T}_{:j}) \quad (18)$$

where $\mathbf{T}_{i:}$ and $\mathbf{T}_{:j}$ represent the ith row and the jth column of the matrix $\mathbf{T}$, respectively.

Since the pixels in one superpixel block tend to be from the same class, a pixel can be represented by its corresponding superpixel block. We employ the SMD between $\mathbf{B}_i$ and $\mathbf{B}_j$ to represent the similarity between pixels $\mathbf{x}_i$ and $\mathbf{x}_j$. We define the similarity between $\mathbf{x}_i$ and $\mathbf{x}_j$ as follows:

$$\mathbf{M}_{ij} = \exp(-d_{\mathrm{smd}}(\mathbf{B}_i, \mathbf{B}_j)). \quad (19)$$

Our goal is to let similar blocks have similar representation coefficients. To approach this goal, we create the following regularizer $B(\mathbf{W})$:

$$B(\mathbf{W}) = \frac{1}{2}\sum_{i,j}\left\|\mathbf{w}_i - \mathbf{w}_j\right\|^2\mathbf{M}_{ij} \quad (20)$$

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where $\mathbf{w}_i$ and $\mathbf{w}_j$ represent the coefficients corresponding to $\mathbf{x}_i$ and $\mathbf{x}_j$, respectively.

We add (20) into (5), then we can obtain the SPA-G model as follows:

$$\min_{\mathbf{W}} \frac{1}{2}\|\mathbf{X}-\mathbf{X}\mathbf{W}\|_F^2+\gamma_1\|\mathbf{W}\|_1+\gamma_2\sum_{i,j}\left\|\mathbf{w}_i-\mathbf{w}_j\right\|^2\mathbf{M}_{ij}$$
$$\text{s.t. } \operatorname{diag}(\mathbf{W})=0,\quad \mathbf{W}\geq 0 \tag{21}$$

where $\gamma_1 > 0$ is used to balance the sparsity and the reconstruction error, and $\gamma_2 > 0$ is a parameter to control the weight of the regularizer. The objective function (21) is written as

$$\min_{\mathbf{W}} \frac{1}{2}\|\mathbf{X}-\mathbf{X}\mathbf{W}\|_F^2+\gamma_1\|\mathbf{W}\|_1+\gamma_2\operatorname{tr}(\mathbf{W}\mathbf{L}\mathbf{W}^{\mathrm{T}})$$
$$\text{s.t. } \operatorname{diag}(\mathbf{W})=0,\quad \mathbf{W}\geq 0 \tag{22}$$

where $\mathbf{L}=\mathbf{D}-\mathbf{M}$ is the Laplacian matrix, and $\mathbf{D}_{ii}=\sum_j \mathbf{M}_{ij}$ is the diagonal degree matrix.

We use the ADMAP [45] method to solve (22). According to [45], we introduce auxiliary variable $\mathbf{P}$ to separate (22), then the objective function can be formulated as

$$\min_{\mathbf{W},\mathbf{J}} \frac{1}{2}\|\mathbf{X}-\mathbf{X}\mathbf{P}\|_F^2+\gamma_1\|\mathbf{W}\|_1+\gamma_2\operatorname{tr}(\mathbf{P}\mathbf{L}\mathbf{P}^{\mathrm{T}})$$
$$\text{s.t. } \mathbf{P}=\mathbf{W}-\operatorname{diag}(\mathbf{W}),\quad \mathbf{W}\geq 0. \tag{23}$$

Then, the augmented Lagrangian function of (23) is as follows:

$$L(\mathbf{W},\mathbf{P},\mathbf{\Lambda},\mu)=\frac{1}{2}\|\mathbf{X}-\mathbf{X}\mathbf{P}\|_F^2+\gamma_1\|\mathbf{W}\|_1+\gamma_2\operatorname{tr}\left(\mathbf{P}\mathbf{L}\mathbf{P}^{\mathrm{T}}\right)+\frac{\mu}{2}\|\mathbf{P}-\mathbf{W}+\operatorname{diag}(\mathbf{W})\|_F^2+\operatorname{tr}(\mathbf{\Lambda}^{\mathrm{T}}(\mathbf{P}-\mathbf{W}+\operatorname{diag}(\mathbf{W}))) \tag{24}$$

where μ is a numerical parameter, and $\mathbf{\Lambda}$ is the Lagrange multiplier.

To minimize (24), we update variables $\mathbf{W}$, $\mathbf{P}$, and $\mathbf{\Lambda}$ alternately. Detailed updating steps are as follows.

1) Fix other variables and update $\mathbf{P}$

$$\mathbf{P}=\arg\min_{\mathbf{P}} \frac{1}{2}\|\mathbf{X}-\mathbf{X}\mathbf{P}\|_F^2+\gamma_2\operatorname{tr}\left(\mathbf{P}\mathbf{L}\mathbf{P}^{\mathrm{T}}\right)+\frac{\mu}{2}\|\mathbf{P}-\mathbf{W}\|_F^2+\operatorname{tr}(\mathbf{\Lambda}^{\mathrm{T}}(\mathbf{P}-\mathbf{W})). \tag{25}$$

We set the partial derivative of (25) with respect to $\mathbf{P}$ to zero, and we can obtain the following equation:

$$\mathbf{P}(2\gamma_2\mathbf{L})+\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}+\mu\mathbf{I}\right)\mathbf{P}=\mathbf{X}^{\mathrm{T}}\mathbf{X}+\mu\mathbf{W}-\mathbf{\Lambda}. \tag{26}$$

We first perform the following matrix decompositions:

$$2\gamma_2\mathbf{L}=\mathbf{Q_1}\mathbf{R_1}\mathbf{Q_1^{\mathrm{T}}},\quad \mathbf{X}^{\mathrm{T}}\mathbf{X}+\mu\mathbf{I}=\mathbf{Q_2}\mathbf{R_2}\mathbf{Q_2^{T}}. \tag{27}$$

Then, we obtain the solution of (26)

$$\mathbf{P}=\mathbf{Q}_2\tilde{\mathbf{P}}\mathbf{Q}_1^{\mathrm{T}} \tag{28}$$

where $\tilde{\mathbf{P}}(i,j)=(\mathbf{G}(i,j)/\mathbf{R}_2(i,i)+\mathbf{R}_1(j,j))$, $\mathbf{G}=\mathbf{Q_2^T}\mathbf{B}\mathbf{Q_1}$, and $\mathbf{B}=\mathbf{X}^{\mathrm{T}}\mathbf{X}+\mu\mathbf{W}-\mathbf{\Lambda}$.

2) Fix other variables and update $\mathbf{W}$

$$\mathbf{W}=\arg\min_{\mathbf{W}} \gamma_1\|\mathbf{W}\|_1+\frac{\mu}{2}\|\mathbf{P}-\mathbf{W}+\operatorname{diag}(\mathbf{W})\|_F^2+\operatorname{tr}(\mathbf{\Lambda}^{\mathrm{T}}(\mathbf{P}-\mathbf{W}+\operatorname{diag}(\mathbf{W}))). \tag{29}$$

The solution of (29) is

$$\mathbf{W}=\tilde{\mathbf{W}}-\operatorname{diag}(\tilde{\mathbf{W}}) \tag{30}$$

where each entry of $\tilde{\mathbf{W}}$ is computed by

$$\tilde{\mathbf{W}}_{ij}=\max\left(\Theta_{((\gamma_1\mathbf{I})\mu^{-1})}\left((\mathbf{P}+\frac{\mathbf{\Lambda}}{u})_{ij}\right),0\right) \tag{31}$$

where $\Theta_{(\varepsilon)}(t)=\max(|t|-\varepsilon,0)\operatorname{sgn}(t)$ is a soft thresholding function.

3) Fix other variables and update $\mathbf{\Lambda}$

$$\mathbf{\Lambda}=\mathbf{\Lambda}+\mu(\mathbf{P}-\mathbf{W}). \tag{32}$$

The procedure of solving (21) is described in Algorithm 2.

The process of constructing SPA-G is similar to that of constructing SPE-G. For the given data $\mathbf{X}$, we use Algorithm 2 to obtain the coefficient matrix $\hat{\mathbf{W}}_2$ of $\mathbf{X}$. Then, we utilize (33) to obtain the symmetric matrix $\mathbf{W}^2$. Then, the matrix $\mathbf{W}^2$ is regarded as the adjacency matrix of SPA-G

$$\mathbf{W}^2=\left(\hat{\mathbf{W}}_2+\hat{\mathbf{W}}_2^{\mathrm{T}}\right)/2. \tag{33}$$

Algorithm 2 Solving Problem (21) by ADMAP

Input: data points $\mathbf{X}=\{\mathbf{x}_1,\mathbf{x}_2,\cdots\mathbf{x}_n\}\in\mathbb{R}^{d\times n}$, parameters $\gamma_1>0$, $\gamma_2>0$, the number of superpixel blocks: $N>0$.
1: **Initialization:** $\mathbf{W}=\mathbf{0}$, $\mathbf{P}=\mathbf{0}$, $\mathbf{\Lambda}=\mathbf{0}$, $\mu^{(0)}=0.1$, $\mu_{\max}=10^{10}$, $\rho_0=1.1$, $\varepsilon_1=10^{-4}$, $\varepsilon_2=10^{-4}$, $\eta=\|\mathbf{X}\|_2^2+1$, $k=0$.
2: Perform PCA for spectral features.
3: Segment the image into N superpixel blocks on the first principal component. Compute the matrix $\mathbf{M}$ by using (19).
4: Extract the 2-D Gabor features from the first three principal components. Feed $\mathbf{X}$ with the 2-D Gabor features and the matrix $\mathbf{M}$ into (21).
5: **While** $\|\mathbf{X}-\mathbf{X}\mathbf{W}\|_F/\|\mathbf{X}\|_F\geq\varepsilon_1$ or $\mu_k\max(\sqrt{\eta}\|\mathbf{W}_k-\mathbf{W}_{k-1}\|_F,\|\mathbf{P}_k-\mathbf{P}_{k-1}\|_F)/\|\mathbf{X}\|_F\geq\varepsilon_2$ **do**
6: Update $\mathbf{P}_{k+1}$ by (27) and (28);
7: Update $\mathbf{W}_{k+1}$ by (30) and (31);
8: Update $\mathbf{\Lambda}_{k+1}$ by (32);
9: Update $\mu^{(k+1)}$: $\mu^{(k+1)}=\min(\mu_{\max},\rho\mu^{(k)})$, where
$$\rho=\begin{cases}\rho_0, \text{if } \mu^{(k)}\max(\sqrt{\eta}\|\mathbf{W}-\mathbf{W}_{k-1}\|_F,\\ \quad\|\mathbf{P}-\mathbf{P}_{k-1}\|_F)/\|\mathbf{X}\|_F<\varepsilon_2\\ 1, \text{otherwise.}\end{cases}$$
10: Update k: $k\leftarrow k+1$;
11: **End while**
12: **Output:** The coefficient matrix $\mathbf{W}$.

C. DSRG-CP

Traditional graph-based methods only construct one graph for label propagation. Once the graph is obtained, the labels are propagated directly from labeled samples to all unlabeled samples via the graph. When the number of labeled samples is very small, the results of label propagation are generally not reliable. Spectral view and spatial view are independent

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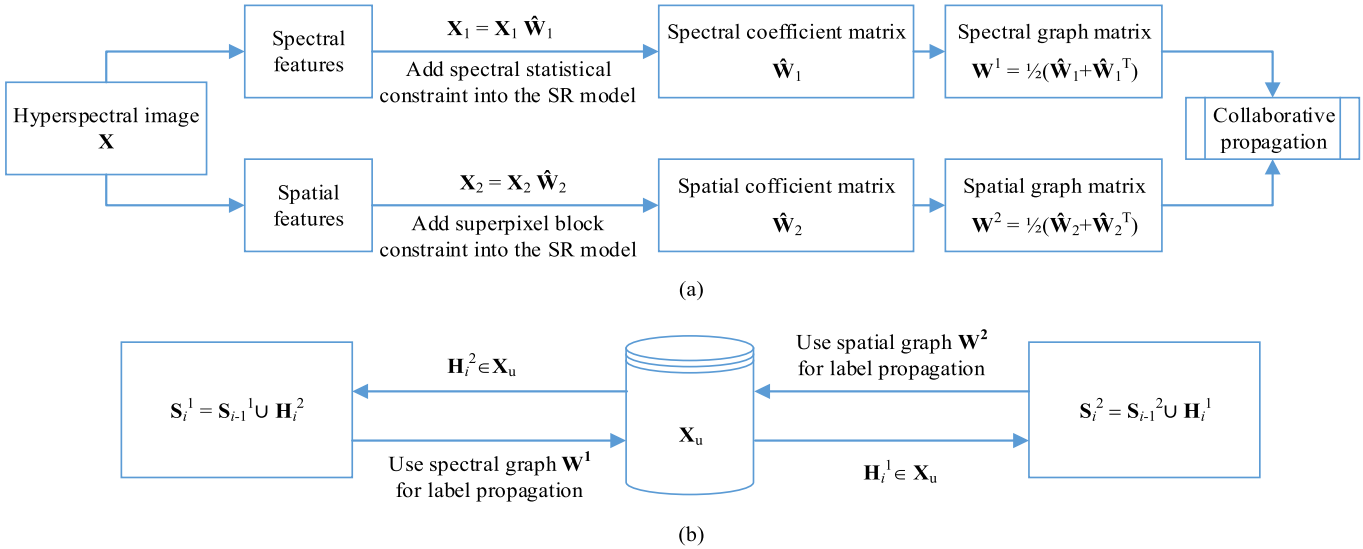


Fig. 1. Flowchart of DSRG-CP. (a) Overall flowchart. (b) Flowchart of copropagation. First, we construct two SR models on spectral and spatial views. Second, we compute the coefficient matrices $\hat{\mathbf{W}}_1$ and $\hat{\mathbf{W}}_2$ by solving two SR models, and then we obtain the adjacency matrices $\mathbf{W}^1$ and $\mathbf{W}^2$ for the spectral and spatial graphs. Third, we perform copropagation on two graphs. In each iteration, two graphs will generate the results of label propagation, the samples with high classification confidence will be added into anther graph as labeled samples for the next propagation. At last, we predict the labels of unlabeled samples by fusing the results of two graphs. (b) Process of copropagation. The symbol $\mathbf{S}_i$ represents labeled samples and $\mathbf{S}_0$ is initial labeled samples, the symbol $\mathbf{X}_u$ is the unlabeled samples, and the symbol $\mathbf{H}_i$ represents the samples with high classification confidence.

of each other, which allows us to use SPE-G and SPA-G for label propagation in the manner of collaboration.

In DSRG-CP, there are two views to describe the relationship among samples, i.e., one view is the spectral graph; the other view is the spatial graph. The main idea of DSRG-CP is to use dual graphs for label propagation in the manner of collaboration. During the iteration of copropagation, only the samples with high classification confidence will be assigned with pseudolabels. Besides, these newly labeled samples from one graph will be added into another graph for the next propagation. The flowchart of DSRG-CP is illustrated in Fig. 1.

The main procedures of DSRG-CP are as follows. First, we construct SPE-G and SPA-G on spectral and spatial views, respectively. Second, the labels are propagated from labeled samples to unlabeled samples for SPE-G and SPA-G, and some newly labeled samples with high confidence are added into another graph. Third, the labels of initial labeled and newly labeled samples are propagated to the remaining unlabeled samples. Fourth, we repeat the second and the third steps several times until no newly labeled samples can be added into another graph. At last, we use (34) to predict the final class label for each sample $\mathbf{x}$

$$y_i = \arg\max_{j=1,2\ldots c} (\alpha \mathbf{F}^1(i,j) + (1-\alpha)\mathbf{F}^2(i,j)) \tag{34}$$

where $\mathbf{F}^1$ and $\mathbf{F}^2$ represent the class probability matrices from the spectral graph and the spatial graph, respectively. The form of F is shown as (3), and each entry of $\mathbf{F}$ indicates the probability of the sample $\mathbf{x}_i$ belonging to the class Ω_j. The parameter α is used to balance the significance between SPE-G and SPA-G.

The main steps of DSRG-CP are described in Algorithm 3.

D. Complexity Analysis of DSRG-CP

The overall computational complexity of DSRG-CP consists of three parts, i.e., computing the SPE-G, computing the SPA-G, and performing copropagation.

The main process of computing the SPE-G is updating three variables $\mathbf{P}$, $\mathbf{W}$, and $\mathbf{\Lambda}$ alternatively. Formula (12) shows the process of updating $\mathbf{P}$, whose complexity is $O(n^3)$, where n is the number of samples. The complexity of updating $\mathbf{W}$ and $\mathbf{\Lambda}$ is both $O(n^2)$. So, the complexity of computing the SPE-G is $O(\tau_1 n^3)$, where τ_1 represents the number of iterations in Algorithm 1. The main process of solving SPA-G is similar to that of solving SPE-G. We can obtain that the complexity of computing the SPA-G is $O(\tau_2 n^3)$, where τ_2 is the number of iterations in Algorithm 2. The complexity of label propagation is $O(n^3)$ [30]. Copropagation uses two graphs for label propagation iteratively. So, the complexity of copropagation is $O(2\tau_3 n^3)$, where τ_3 is the number of iterations in Algorithm 3. To sum up, the complexity of DSRG-CP is $O((\tau_1+\tau_2+2\tau_3)n^3)$. Generally, the iterations of three algorithms are about dozens or hundreds of times. Hence, the overall complexity can be regarded as $O(n^3)$.

IV. Experimental Results

A. Hyperspectral Data Sets

Three HSI data sets are used to evaluate the performance of DSRG-CP.

1) The Kennedy Space Center (KSC) data were acquired by the NASA Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor over the KSC, Florida, on March 23, 1996. The original data have 224 spectral bands. We use only 176 bands for experiments because we exclude water absorption bands and bands with low

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Algorithm 3 DSRG-CP

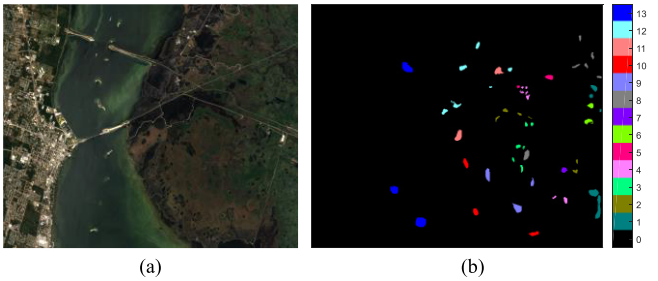
Input: labeled data $\mathbf{X}_l = \{\mathbf{x}_1, \ldots, \mathbf{x}_l\} \in \mathbb{R}^{d\times l}$, unlabeled data $\mathbf{X}_u = \{\mathbf{x}_{l+1}, \cdots \mathbf{x}_{l+u}\} \in \mathbb{R}^{d\times u}$, $\mathbf{X} = \{\mathbf{X}_l, \mathbf{X}_u\} \in \mathbb{R}^{d\times n}$, parameters $\lambda_1 > 0$, $\lambda_2 > 0$; $\gamma_1 > 0$, $\gamma_2 > 0$, the number of superpixel blocks: $N > 0$, the threshold probability π, iteration epoch $k = 0$, the parameter α.

1: Compute the spectral coefficients $\hat{\mathbf{W}}_1^{\mathrm{T}}$ and spatial coefficients $\hat{\mathbf{W}}_2^{\mathrm{T}}$ through Algorithm 1 and Algorithm 2, respectively.
2: Use (17) and (33) to obtain the SPE-G matrix $\mathbf{W}^1$ and SPA-G matrix $\mathbf{W}^2$, respectively.
3: Perform label propagation from $\mathbf{X}_l$ to $\mathbf{X}_u$ on graph $\mathbf{W}^v (v = 1, 2)$, and obtain the probability matrices $\mathbf{F}_k^v (v = 1, 2)$ of $\mathbf{X}_u$ through (3).
4: Obtain the newly labeled samples $\mathbf{H}_k^v$ $(v = 1, 2)$, where each sample $\mathbf{x}$ in $\mathbf{H}_k^v$ is satisfied with $\max(\mathbf{F}_k^v(\mathbf{x})) \geq \pi$. // $\mathbf{F}(\mathbf{x})$ represents the predicted class probability vector of $\mathbf{x}$, and $\max(\mathbf{F}(\mathbf{x}))$ is the maximum in $\mathbf{F}(\mathbf{x})$.
5: Get labeled sample set $\mathbf{S}_k^v$ $(v = 1, 2)$: $\mathbf{S}_k^v = \mathbf{X}_l \cup \mathbf{H}_k^v$.
6: **While** $\mathbf{H}_k^1 \neq \emptyset$ or $\mathbf{H}_k^2 \neq \emptyset$
7: $k = k + 1$.
8: Perform label propagation from labeled set $\mathbf{S}_k^{3-v} \cap (\mathbf{X}_u - \mathbf{S}_k^v)$ to unlabeled set $\mathbf{X}_u - \mathbf{S}_k^v$ on graph $\mathbf{W}^v$, and obtain the probability matrices $\mathbf{F}_k^v$ of $\mathbf{X}_u - \mathbf{S}_k^v$;
9: Obtain the newly labeled samples set $\mathbf{H}_k^v$, where each sample $\mathbf{x}$ in $\mathbf{H}_k^v$ is satisfied with $\max(\mathbf{F}_k^v(\mathbf{x})) \geq \pi$.
10: Update $\mathbf{S}_k^v$: $\mathbf{S}_{k+1}^v = \mathbf{S}_k^v \cup \mathbf{H}_k^v$.
11: **End while**
12: Obtain probability matrices $\mathbf{F}^v$ corresponding to $\mathbf{S}_k^v$.
13: **Output:** the predicted labels of unlabeled samples $\mathbf{x}_i\, y_i = \arg \max_{j=1,2\ldots c} (\alpha \mathbf{F}^1(i, j) + (1 - \alpha)\mathbf{F}^2(i, j))$ $(i = l + 1, \ldots, l + u)$.

signal-to-noises. The data set contains 13 classes with a size of 512. × 614 pixels, and the spatial resolution is 18 m. There are a total of 5211 labeled pixels.

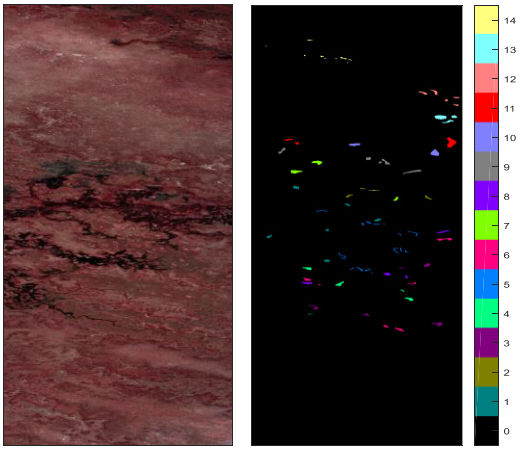
2) The Botswana (BOT) data were obtained by the NASA Earth Observing-1 satellite over the Okavango Delta, Botswana, on May 11, 2001. The original data have 242 bands. After removing uncalibrated and noisy bands, the remaining 145 bands are used in the experiments. The BOT data set is of 1476. × 256 pixels in size and has a spatial resolution of 30 m. There are a total of 3248 labeled pixels covering 14 classes.
3) The Salinas Valley (SV) data were obtained by the AVIRIS sensor in 1998, which captured the land over Salinas Valley in California. This data set has 512. × 217 pixels with a spatial resolution of 3.7 m. The original data have 224 spectral bands. We discard 20 water absorption bands, and the remaining spectral bands are used in our experiments. A total of 54 129 pixels covering 16 classes are labeled in reference map.

The three-band pseudocolor images of the three HSI data sets and their corresponding reference maps are shown in Figs. 2–4.



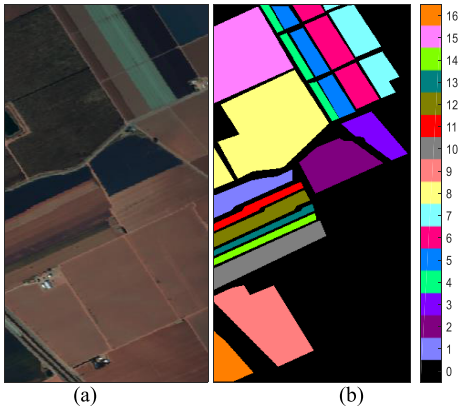
(a) (b)

Fig. 2. (a) Three-band pseudocolor composite image of the KSC data set and (b) its corresponding reference map.



(a) (b)

Fig. 3. (a) Three-band pseudocolor composite image of the BOT data set and (b) its corresponding reference map.



(a) (b)

Fig. 4. (a) Three-band pseudocolor image of the SV data set and (b) its corresponding reference map.

B. Experimental Setup

In Algorithm 2, the 2-D Gabor filter is adopted to extract the spatial features, and the parameters of Gabor filter are set the same as those in [52]. For each image used in our experiments, we used simple linear iterative clustering (SLIC) [50] to segment the image into 1500 superpixel blocks, i.e., $N = 1500$ in Algorithm 3. In Algorithm 3, the parameter α is set to 0.4 according to several trials, and the parameter π is set to 0.6. In the parameter sensitivity analysis stage,

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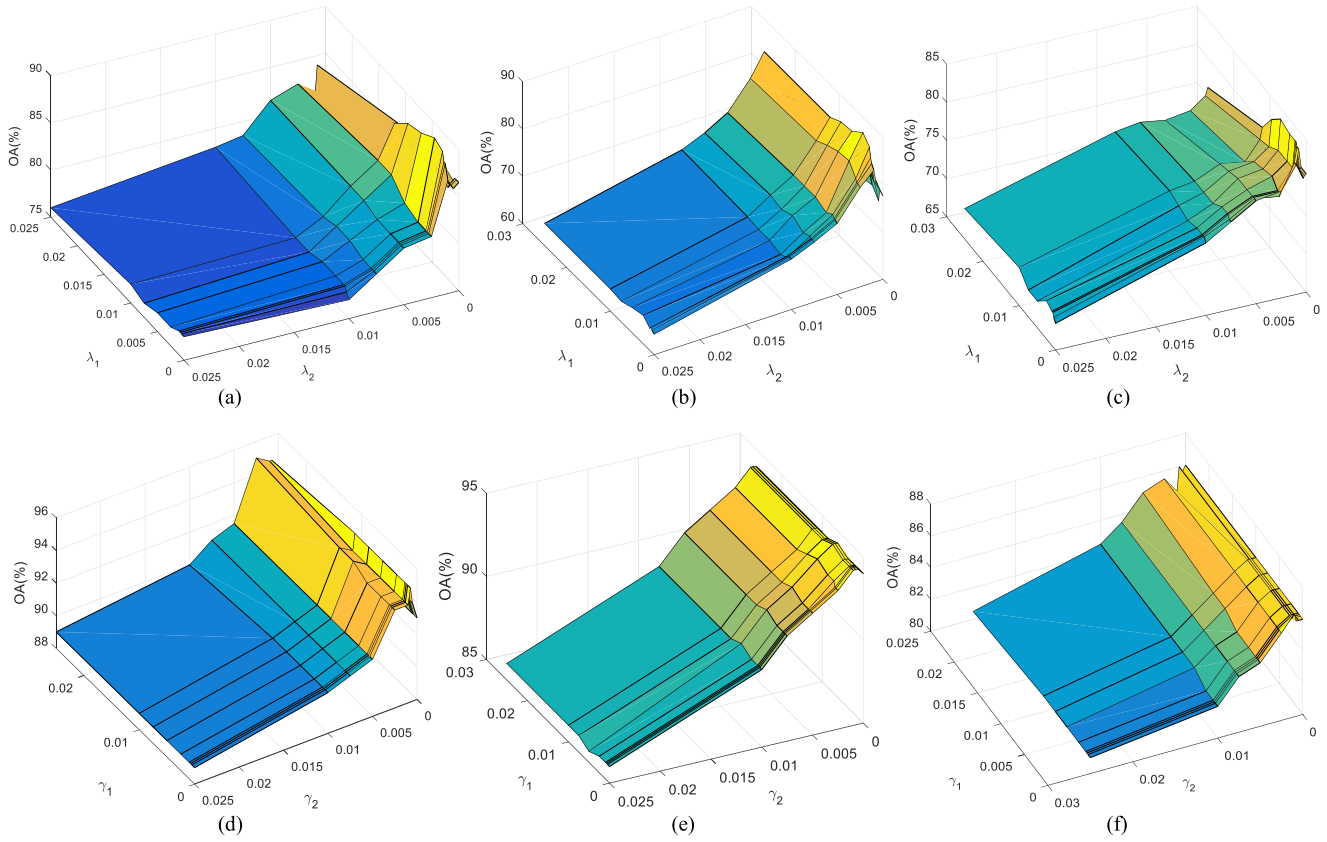


Fig. 5. Classification accuracies of two graphs under different pairs of parameters on three images. (a)–(c) Influence of pairs of parameters λ_1 and λ_2 in SPE-G on the accuracies of KSC, BOT, and SV data, respectively. (d)–(f) Influence of pairs of parameters γ_1 and γ_2 in SPA-G on the accuracies of KSC, BOT, and SV data, respectively.

we will analyze the influence of parameter N and α on the classification performance.

In our experiments, we use three measures, including overall accuracy (OA), average accuracy (AA), and kappa coefficient, to analyze the classification performance of different approaches. For each method, ten runs of experiments are executed on each data set to obtain the average results.

In SPE-G and SPA-G, they both have one pair of parameters (λ_1 and λ_2 for SPE-G, γ_1 and γ_2 for SPA-G) need to be set. Since SPE-G and SPA-G are independent of each other, we only need to determine these two pairs of parameters separately. We randomly select 20 samples for each class to form the training set. Following Shao *et al.* [23], [26], the leave-one-out (LOO) method is applied to evaluate the performance. In detail, given a training set of n samples, for each pair of parameters, the LOO-based validation is performed with n iterations, such that in each iteration, the classification method is trained with $n-1$ samples and tested on the remaining one sample. At last, the AA of n iterations is treated as the final accuracy under current parameter values. For example, when we verify the SPE-G on KSC data set, there are 260 samples (20 samples per class, a total of 13 classes) used for experiments. For each pair of parameters $.\lambda_1$ and $.\lambda_2$, we perform experiments 260 iterations, such that in each iteration, the classification model is trained with 259 samples and tested on the remaining one sample. After 260 iterations, the AA of 260 iterations is recorded. We adjust the parameter values of λ_1 and λ_2 multiple times to repeat above experiments, and the average accuracies under different pairs of parameters are depicted in Fig. 5(a).

TABLE I
OPTIMAL PARAMETER VALUES FOR DIFFERENT GRAPHS

	λ_1/γ_1	λ_2/γ_2
SPE-G-NoR	$\lambda_1 = 0.0005{\sim}0.005$	$\lambda_2 = 0$
SPE-G	$\lambda_1 = 0.0005{\sim}0.005$	$\lambda_2 = 0.0005{\sim}0.001$
SPA-G-NoR	$\gamma_1 = 0.0005{\sim}0.01$	$\gamma_2 = 0$
SPA-G	$\gamma_1 = 0.0005{\sim}0.01$	$\gamma_2 = 0.0005{\sim}0.001$

Fig. 5 shows the classification accuracies of two graphs under different pairs of parameters on three images. Table I shows the optimal parameter values for different graphs, where SPE-G-NoR ($\lambda_2 = 0$) and SPA-G-NoR ($\gamma_2 = 0$) represent SPE-G and SPA-G without regularizations, respectively.

C. Evaluation of the Roles of Different Parts in DSRG-CP

In this section, we perform experiments to evaluate the roles of different parts in DSRG-CP. For each data set, we randomly select 20 samples per class as labeled, and the remaining samples are treated as unlabeled. According to Table I, we set the parameters of compared methods as follows.

SPE-G-NoR ($\lambda_1 = 0.001, \lambda_2 = 0$) : The spectral SR graph without the regularization of spectral statistical information.

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TABLE II
CLASS-SPECIFIC ACCURACIES, OAS, AAS, AND KAPPA COEFFICIENTS (IN %) OBTAINED BY DIFFERENT GRAPHS WHEN APPLIED TO THE KSC IMAGES

Class No./Name	SPE-G-NoR	SPE-G	SPA-G-NoR	SPA-G	DSRG-CP
1/Scrub	87.48 ± 4.27	89.35 ± 4.04	95.14 ± 4.89	98.19 ± 3.96	**99.41 ± 0.64**
2/Willow	81.74 ± 8.56	84.47 ± 9.70	77.47 ± 9.71	93.75 ± 7.76	**99.09 ± 2.05**
3/CP hammock	95.78 ± 6.68	95.22 ± 4.85	94.29 ± 8.32	95.52 ± 8.85	**97.65 ± 4.96**
4/CP/Oak	42.14 ± 11.00	65.33 ± 9.14	95.12 ± 5.43	93.98 ± 9.73	**99.41 ± 1.86**
5/Slash pine	71.07 ± 9.91	75.31 ± 10.08	93.62 ± 9.82	92.71 ± 13.21	**98.86 ± 2.41**
6/Oak/Broadleaf	65.89 ± 14.97	67.61 ± 7.80	**100.00 ± 0.00**	**100.00 ± 0.00**	99.31 ± 2.18
7/Hardwood swamp	97.52 ± 4.06	99.86 ± 0.45	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
8/Graminoid marsh	81.54 ± 7.99	84.54 ± 6.99	81.75 ± 8.25	91.83 ± 6.71	**97.16 ± 4.37**
9/Spartina marsh	89.05 ± 4.84	93.08 ± 4.22	95.78 ± 4.68	97.46 ± 5.67	**99.13 ± 1.56**
10/Cattail marsh	93.51 ± 3.70	92.86 ± 4.92	94.72 ± 9.20	95.39 ± 9.93	**100.00 ± 0.00**
11/Salt marsh	93.99 ± 5.67	92.49 ± 8.25	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
12/Mud flats	76.32 ± 3.92	81.24 ± 4.26	83.60 ± 12.23	91.40 ± 9.48	**99.57 ± 1.36**
13/Water	99.08 ± 1.18	98.22 ± 1.52	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
AA	82.70 ± 1.82	85.13 ± 1.20	93.08 ± 1.56	96.08 ± 1.76	**99.20 ± 0.52**
OA	85.70 ± 1.41	88.18 ± 1.14	92.90 ± 1.53	96.67 ± 1.43	**99.38 ± 0.35**
kappa	84.08 ± 1.55	86.74 ± 1.25	92.53 ± 2.18	96.28 ± 1.61	**99.30 ± 0.40**

TABLE III
CLASS-SPECIFIC ACCURACIES, OAS, AAS, AND KAPPA COEFFICIENTS (IN %) OBTAINED BY DIFFERENT GRAPHS WHEN APPLIED TO THE BOT IMAGES

Class No./Name	SPE-G-NoR	SPE-G	SPA-G-NoR	SPA-G	DSRG-CP
1/Water	99.64 ± 1.13	**100.00 ± 0.00**	85.35 ± 9.41	91.10 ± 9.88	99.52 ± 1.51
2/Hippo grass	99.17 ± 1.83	**100.00 ± 0.00**	95.65 ± 9.21	**100.00 ± 0.00**	**100.00 ± 0.00**
3/Floodplain grass1	98.48 ± 2.34	99.21 ± 1.83	92.52 ± 9.33	98.75 ± 2.64	**99.98 ± 0.06**
4/Floodplain grass2	99.72 ± 0.88	**100.00 ± 0.00**	98.68 ± 2.84	94.39 ± 6.25	**100.00 ± 0.00**
5/Reeds1	81.37 ± 6.36	86.44 ± 3.65	82.67 ± 10.42	91.40 ± 9.89	**99.57 ± 1.37**
6/Riparian	58.86 ± 13.55	74.08 ± 9.20	92.67 ± 5.05	94.37 ± 4.40	**97.08 ± 2.91**
7/Firescar2	97.82 ± 1.46	98.55 ± 2.42	96.80 ± 6.77	98.72 ± 4.05	**99.74 ± 0.81**
8/Island interior	95.08 ± 6.20	99.31 ± 1.47	98.91 ± 2.67	97.69 ± 7.30	**99.72 ± 0.88**
9/Acacia woodlands	86.32 ± 6.19	87.88 ± 6.87	97.64 ± 5.74	**100.00 ± 0.00**	99.99 ± 0.03
10/Acacia shrublands	63.06 ± 9.68	71.23 ± 8.12	97.55 ± 7.76	**100.00 ± 0.00**	**100.00 ± 0.00**
11/Acacia grasslands	96.11 ± 3.36	97.02 ± 2.52	93.87 ± 10.94	98.93 ± 3.39	**99.64 ± 1.13**
12/Short mopane	97.27 ± 2.84	98.92 ± 1.95	98.75 ± 2.96	**100.00 ± 0.00**	99.96 ± 0.13
13/Mixed mopane	76.61 ± 10.23	78.71 ± 9.05	100.00 ± 0.00	**100.00 ± 0.00**	**100.00 ± 0.00**
14/Exposed soils	99.44 ± 1.76	98.70 ± 2.76	91.78 ± 9.47	99.78 ± 0.46	**99.89 ± 0.33**
AA	88.61 ± 2.63	91.93 ± 1.68	94.27 ± 2.08	97.30 ± 1.45	**99.66 ± 0.21**
OA	87.18 ± 2.58	90.51 ± 2.05	93.94 ± 2.18	97.12 ± 1.30	**99.65 ± 0.21**
kappa	85.96 ± 3.02	89.68 ± 2.23	93.53 ± 2.19	96.88 ± 1.41	**99.61 ± 0.23**

SPE-G ($\lambda_1 = 0.001, \lambda_2 = 0.0005$) : The spectral SR graph with the regularization of spectral statistical information.

SPA-G-NoR ($\gamma_1 = 0.005, \gamma_2 = 0$) : The spatial SR graph without the regularization of superpixel block similarity.

SPA-G ($\gamma_1 = 0.005, \gamma_2 = 0.0005$) : The spatial SR graph with the regularization of superpixel block similarity.

DSRG-CP ($\lambda_1 = 0.001, \lambda_2 = 0.0005; \gamma_1 = 0.005, \gamma_2 = 0.0005$) : The dual SR graph with copropagation.

Tables II–IV describe the class-specific accuracies, OAs, AAs, and kappa coefficients of KSC, BOT, and SV data sets, respectively. For the KSC images, Table II shows that SPA-G-NoR, SPA-G, and DSRG-CP obtain the best class-specific accuracies on 4, 4, and 12 classes, respectively. For the BOT images, the results in Table III exhibit that SPE-G, SPA-G, and DSRG-CP achieve the best class-specific accuracies on 3, 5, and 11 classes independently. For the SV images, Table IV demonstrates that SPE-G-NoR, SPE-G, SPA-G-NoR SPA-G, and DSRG-CP obtain the best class-specific accuracies on 5, 4, 1, 2, and 13 classes, respectively. For three data sets, both spectral graph and spectral graph get the best accuracies on some classes, but the accuracies of some other classes are very low. For example, in Table II, the accuracies of classes 4, 6, and 7 obtained by spectral graphs are very low. For both spectral and spatial graphs, the regularizations added into graphs increase the accuracies on most classes. Although the regularizations make the accuracies of some classes decrease, the magnitude of the decrease is small and the OA is increased. One shortcoming of DSRG-CP can be found from Tables II–IV. The SPE-G and SPA-G get 100% class-specific accuracies on some classes, but DSRG-CP cannot hold on. The reason for this phenomenon is that DSRG-CP is influenced by both SPE-G and SPA-G. DSRG-CP utilizes the complementarity of two graphs to

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TABLE IV

CLASS-SPECIFIC ACCURACIES, OAS, AAS, AND KAPPA COEFFICIENTS (IN %) OBTAINED BY DIFFERENT GRAPHS WHEN APPLIED TO THE SV IMAGES

Class No./Name	SPE-G-NoR	SPE-G	SPA-G-NoR	SPA-G	DSRG-CP
1/Brocoli_green_weeds_1	99.05 ± 2.02	99.66 ± 1.09	97.76 ± 5.55	99.09 ± 2.87	**99.95 ± 0.19**
2/Brocoli_green_weeds_2	**100.00 ± 0.00**	**100.00 ± 0.00**	96.80 ± 3.32	93.96 ± 7.97	**100.00 ± 0.00**
3/Fallow	89.10 ± 7.18	97.39 ± 3.84	97.57 ± 4.48	99.33 ± 2.11	**99.98 ± 0.09**
4/Fallow_rough_plow	98.45 ± 3.27	**100.00 ± 0.00**	90.93 ± 9.29	**100.00 ± 0.00**	**100.00 ± 0.00**
5/Fallow_smooth	**98.63 ± 2.32**	98.24 ± 4.52	85.06 ± 6.93	93.32 ± 5.02	97.02 ± 5.32
6/Stubble	**100.00 ± 0.00**	**100.00 ± 0.00**	91.35 ± 7.47	91.43 ± 5.58	98.79 ± 1.54
7/Celery	**100.00 ± 0.00**	**100.00 ± 0.00**	94.51 ± 4.31	98.36 ± 3.29	**100.00 ± 0.00**
8/Grapes_untrained	62.65 ± 13.00	71.27 ± 9.52	69.71 ± 7.69	80.29 ± 6.76	**92.18 ± 4.51**
9/Soil_vinyard_develop	99.80 ± 0.62	99.74 ± 0.81	97.90 ± 2.08	99.69 ± 0.97	**99.94 ± 0.24**
10/Corn_senesced_green_weeds	83.38 ± 6.21	91.62 ± 5.20	86.94 ± 10.56	93.65 ± 9.45	**95.53 ± 4.20**
11Lettuce_romaine_4wk	83.24 ± 11.56	90.27 ± 10.57	94.81 ± 6.87	98.32 ± 4.04	**99.99 ± 0.03**
12/Lettuce_romaine_5wk	**100.00 ± 0.00**	99.38 ± 1.98	91.00 ± 8.39	95.80 ± 4.52	**100.00 ± 0.00**
13Lettuce_romaine_6wk	98.57 ± 4.52	98.57 ± 4.52	89.06 ± 10.83	97.41 ± 6.39	**100.00 ± 0.00**
14/Lettuce_romaine_7wk	88.64 ± 8.58	89.62 ± 10.95	93.78 ± 9.12	95.63 ± 9.34	**99.89 ± 0.32**
15/Vinyard_untrained	66.53 ± 14.85	72.27 ± 7.92	90.20 ± 5.71	89.59 ± 6.24	**95.25 ± 3.34**
16/Vinyard_vertical_trellis	92.95 ± 9.89	95.98 ± 5.91	**100.00 ± 0.00**	**100.00 ± 0.00**	98.89 ± 2.93
AA	91.31 ± 0.97	93.90 ± 1.77	91.71 ± 1.83	95.28 ± 1.98	**98.60 ± 0.41**
OA	84.98 ± 1.34	88.80 ± 2.65	88.12 ± 1.98	91.82 ± 2.44	**97.16 ± 0.52**
kappa	83.25 ± 1.49	87.56 ± 2.95	86.85 ± 2.18	90.95 ± 2.68	**96.83 ± 0.60**

TABLE V

OVERALL ACCURACIES OF DIFFERENT GRAPHS WITH DIFFERENT NUMBERS OF LABELED PIXELS WHEN APPLIED TO THE KSC IMAGES

Method	$L = 3$	$L = 5$	$L = 10$	$L = 15$	$L = 20$
GK-graph	79.17 ± 3.34	82.06 ± 2.28	84.31 ± 1.57	85.02 ± 0.95	85.49 ± 0.87
LLR-graph	79.67 ± 3.14	82.67 ± 2.87	84.92 ± 1.64	85.49 ± 1.12	85.91 ± 0.98
LLE-graph	73.45 ± 2.84	78.21 ± 1.75	82.33 ± 1.26	83.06 ± 0.94	83.62 ± 0.76
PCSSR-graph	81.03 ± 2.89	84.25 ± 1.82	87.21 ± 1.34	88.05 ± 0.75	88.59 ± 0.58
SCSSR-graph	84.78 ± 2.91	86.21 ± 1.87	88.68 ± 1.05	90.14 ± 0.84	92.43 ± 0.43
DSRG	85.87 ± 1.58	88.70 ± 0.66	92.54 ± 0.71	94.17 ± 0.64	95.78 ± 0.29
DSRG-CP-region	88.32 ± 1.87	92.74 ± 0.75	97.58 ± 0.84	98.16 ± 0.65	98.54 ± 0.35
DSRG-CP	89.53 ± 1.51	93.65 ± 0.78	98.35 ± 0.71	99.06 ± 0.70	99.38 ± 0.35

increase OA, which naturally leads to the decrease of some class-specific accuracies.

Tables II–IV reveal several findings. First, compared with other graphs, DSRG-CP obtains the best classification results on AAs, OAs, and kappa coefficients for three HSIs. Second, the spatial graphs (SPA-G and SPA-G-NoR) achieve better classification accuracy than spectral graphs (SPE-G and SPE-G-NoR). Third, the regularization of the spectral statistical information in SPE-G plays an important role in increasing the accuracy, which can be viewed from the comparison results between SPE-G and SPE-G-NoR. Fourth, SPA-G shows better classification performance than SPA-G-NoR, which benefits from the regularizer of the superpixel block.

D. Comparison With Other Graph-Based Methods

In this section, we compare DSRG-CP with other graph-based methods. These methods include locally linear embedding-based graph (LLE-graph) [22], probabilistic class structure regularized sparse representation graph (PCSSR-graph) [23], spatial and class structure regularized sparse representation graph (SCSSR-graph) [26], GK-graph [30], and LLR-graph [32], where PCSSR-graph and SCSSR-graph are SR-based graphs. For extensive analysis, the number of labeled samples per class (L) is set to different values. Specifically, L is set to 3, 5, 10, 15, and 20. To investigate the classification performance of DSRG-CP trained on predefined regions, we randomly select different continuous regions as labeled samples for training, and this method is called DSRG-CP-region. In addition, to demonstrate the role of copropagation in DSRG-CP, we add DSRG for comparison. DSRG method only uses two graphs for label propagation once, and the results of two graphs are combined to predict the labels of unlabeled samples. Tables V–VII show the average OAs of KSC, BOT, and SV HSIs, respectively. The classification maps of different approaches on three HSIs are illustrated in Figs. 6–8.

Tables V–VII reveal that DSRG-CP and DRG-CP-region achieve the best and second-best classification accuracies on three HSI data set, no matter how many samples per class are labeled. The OA difference between DSRG-CP and DSRG-CP-region is small, which indicates that the proposed method is not sensitive to the selected training samples. Two reasons can explain this result. One is that the construction of SR-based graph is unsupervised, which does not require labeled samples. The other is that the superpixel method can obtain the structure information without label information. The classification performance of DSRG-CP is significantly better than SCSSR-graph on KSC and SV data sets, and slightly better than SCSRR-graph on BOT data set. The LLE-graph

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TABLE VI

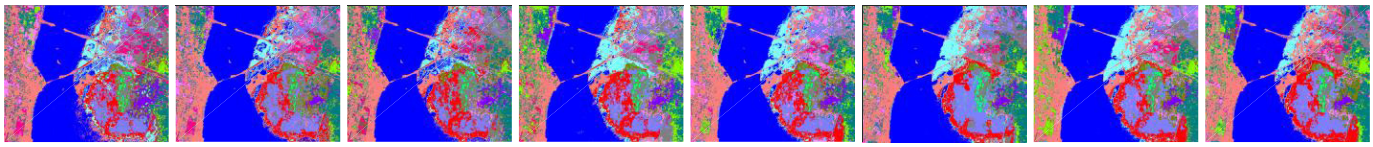
OVERALL ACCURACIES OF DIFFERENT GRAPHS WITH DIFFERENT NUMBERS OF LABELED PIXELS WHEN APPLIED TO THE BOT IMAGES

Method	$L = 3$	$L = 5$	$L = 10$	$L = 15$	$L = 20$
GK-graph	88.96 ± 3.17	90.85 ± 3.02	92.51 ± 2.98	93.22 ± 1.27	93.74 ± 0.82
LLR-graph	89.11 ± 3.87	91.17 ± 3.89	93.05 ± 2.85	94.36 ± 1.75	94.15 ± 1.18
LLE-graph	87.76 ± 3.24	90.53 ± 2.15	93.81 ± 2.05	95.32 ± 1.14	95.82 ± 0.92
PCSSR-graph	91.03 ± 2.96	92.46 ± 1.87	94.88 ± 1.51	95.74 ± 1.22	96.32 ± 0.85
SCSSR-graph	92.34 ± 2.76	93.76 ± 1.85	95.73 ± 1.18	97.16 ± 0.94	97.89 ± 0.76
DSRG	92.39 ± 2.21	93.85 ± 1.58	96.23 ± 1.25	97.74 ± 0.84	98.05 ± 0.69
DSRG-CP-region	92.45 ± 2.35	93.97 ± 1.69	97.96 ± 1.44	98.58 ± 0.74	98.87 ± 0.57
DSRG-CP	92.84 ± 2.04	94.34 ± 1.56	98.23 ± 1.22	99.25 ± 0.40	99.65 ± 0.21

TABLE VII

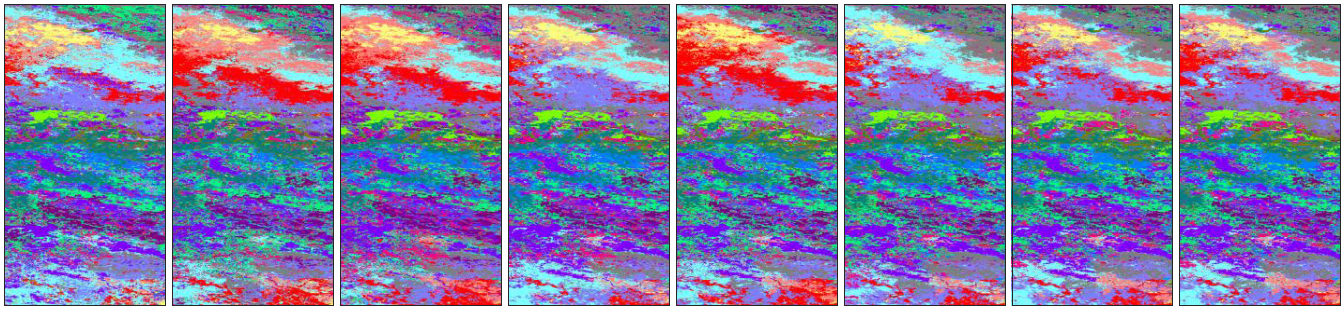
OVERALL ACCURACIES OF DIFFERENT GRAPHS WITH DIFFERENT NUMBERS OF LABELED PIXELS WHEN APPLIED TO THE SV IMAGES

Method	$L = 3$	$L = 5$	$L = 10$	$L = 15$	$L = 20$
GK-graph	79.52 ± 2.87	82.61 ± 2.24	84.59 ± 1.85	86.25 ± 1.40	87.35 ± 0.78
LLR-graph	80.25 ± 2.51	83.45 ± 1.75	85.51 ± 1.89	87.34 ± 1.24	89.84 ± 0.51
LLE-graph	76.42 ± 1.87	81.87 ± 1.34	84.94 ± 1.51	86.34 ± 1.10	87.15 ± 0.72
PCSSR-graph	83.55 ± 1.91	86.51 ± 1.26	88.87 ± 0.98	89.45 ± 0.77	90.45 ± 0.66
SCSSR-graph	85.12 ± 1.76	87.74 ± 1.31	89.84 ± 1.05	91.18 ± 0.89	92.03 ± 0.54
DSRG	86.54 ± 1.27	89.21 ± 1.35	91.12 ± 1.05	92.95 ± 0.74	93.92 ± 0.55
DSRG-CP-region	88.25 ± 1.35	91.35 ± 1.17	93.65 ± 1.21	95.74 ± 0.95	96.51 ± 0.68
DSRG-CP	89.66 ± 1.10	92.36 ± 1.04	94.22 ± 0.82	96.27 ± 0.80	97.16 ± 0.52



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 6. Classification maps of different methods on KSC images when 20 samples per class are labeled. (a) LLE-graph (OA = 83.59%). (b)GK-graph (OA = 85.61%). (c) LLR-graph (OA = 85.97%). (d) PCSSR-graph (OA = 88.65%). (e) SCSSR-graph (OA = 92.78%). (f) DSRG (OA = 95.81%). (g) DSRG-CP-region (OA = 98.55%). (h) DSRG-CP (OA = 99.36%).



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 7. Classification maps of different methods on BOT images when 20 samples per class are labeled. (a) GK-graph (OA = 93.75%). (b) LLR-graph (OA = 94.22%). (c) LLE-graph (OA = 95.86%). (d) PCSSR-graph (OA = 96.37%). (e) SCSSR-graph (OA = 97.88%). (f) DSRG (OA = 98.06%). (g) DSRG-CP-region (OA = 98.85%). (h) DSRG-CP (OA = 99.64%).

is susceptible to the number of labeled samples. When the number of labeled samples is very limited, the average OAs of the LLE-graph are much lower than the GK-graph and LLR-graph. However, when the number of labeled samples is large, the average OAs of the LLE-graph are close to the GK-graph and LLR-graph on three data sets. The average OAs of PCSSR-graph are lower than that of SCSSR-graph, because the latter adds spatial information constraint into the former. In addition, an important finding is that SR-based graphs (PCSSR-graph, SCSSR-graph, and DSRG-CP) consistently outperform traditional graphs (GK-graph, LLR-graph, and LLE-graph), which benefits from the superiority of SR. Compared with DSRG, the classification performance of DSRG-CP increases obviously, which demonstrates that copropagation plays a very important role in the proposed method.

The computational costs of different graphs on three data sets are reported in Table VIII. The results in Table VIII show that GK-graph needs the least time cost and the

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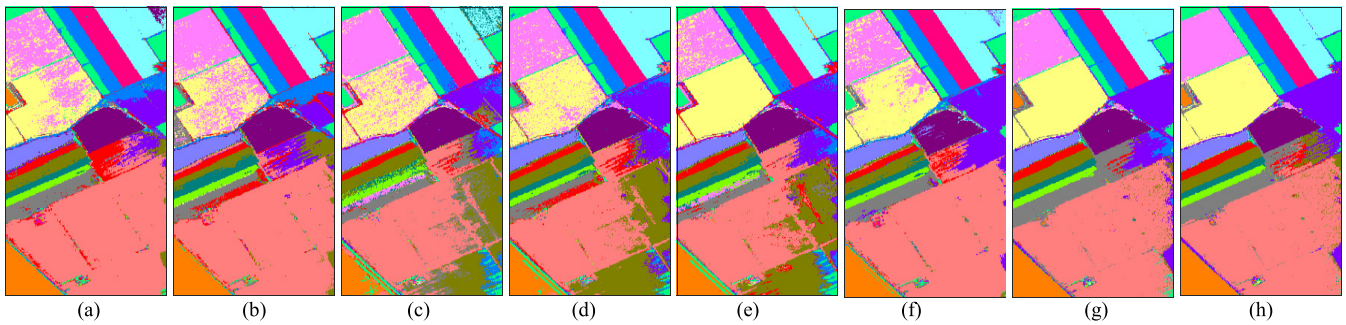


Fig. 8. Classification maps of different methods on SV images when 20 samples per class are labeled. (a) LLE-graph (OA = 87.16%). (b) GK-graph (OA = 87.39%). (c) LLR-graph (OA = 89.86%). (d) PCSSR-graph (OA = 90.55%). (e) SCSSR-graph (OA = 92.18%). (f) DSRG (OA = 93.95%). (g) DSRG-CP-region (OA = 96.53 %). (h) DSRG-CP (OA = 97.23%).

TABLE VIII
OVERALL COMPUTATIONAL TIME OF DIFFERENT GRAPHS ON THREE IMAGES (SECONDS)

Data Set	GK-graph	LLR-graph	LLE-graph	PCSSR-graph	SCSSR-graph	DSRG-CP
KSC	4.7	6.2	24.6	148.6	435.1	815.7
BOT	2.3	3.1	11.5	87.6	268.5	319.6
SV	95.8	123.6	214.2	284.9	615.7	1046.5

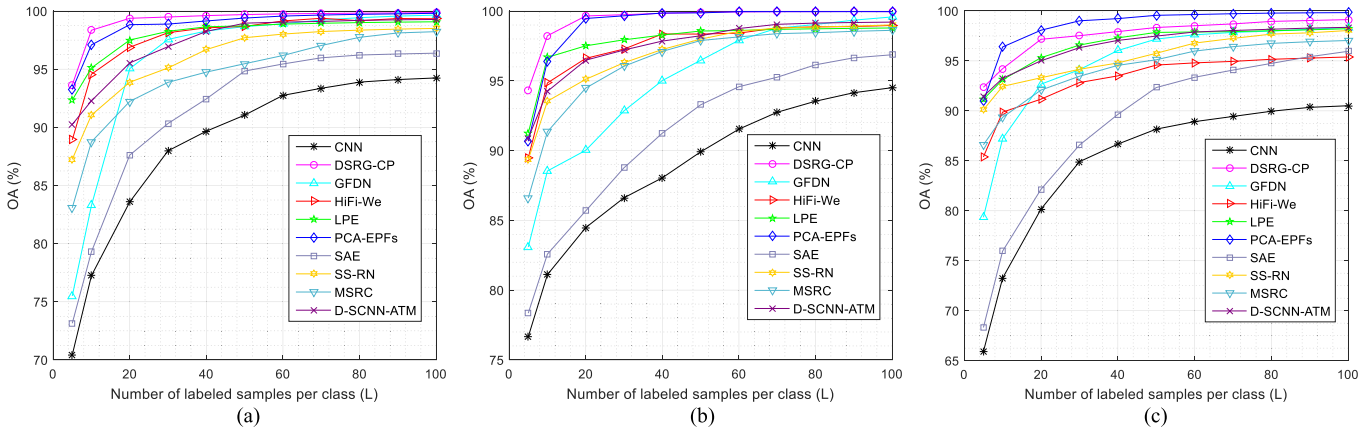


Fig. 9. Average OAs of comparison methods when L equals to different values. (a) KSC. (b) BOT. (c) SV.

proposed DSRG-CP needs the most time cost. DSRG-CP constructs two SR graphs and uses these two graphs for label propagation iteratively, which increase the computational time much. This is a drawback of DSRG-CP, but it is acceptable because the proposed method can significantly improve the classification accuracy at the cost of additional time.

E. Comparison With State-of-the-Art Approaches

Recently, deep learning-based approaches have obtained extensive attention for HSI classification. In this section, we compare DSRG-CP with deep learning-based approaches and other state-of-the-art methods. These methods include Gabor filtering-based deep network (GFDN) [52], convolutional neural network (CNN) [53], stacked autoencoders (SAEs) [54], ensemble method based on hierarchical guidance filtering and weighted voting (HiFi-We) [55], label propagation ensemble (LPE) [33], PCA-based edge preserving features (PCA-EPFs) [56], spatial-spectral relation network (SS-RN) [57], multiobjective-based SR classifier (MSRC) [58], and dual-path siamese CNN combined with adversarial training and mixup (D-SCNN-ATM) [59], where GFDN, CNN, SAE, SS-RN, and D-SCNN-ATM are deep learning-based methods. To make comprehensive comparison with deep learning-based methods, the number of labeled samples per class (L) is set to 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100. Fig. 9 describes the OA changes of different approaches with different numbers of labeled samples, where the OAs are obtained by computing the average results of ten runs of experiments.

Figs. 10–12 visually show the classification maps of comparison methods on KSC, BOT, and SV data sets when L equals to 5. In addition, the class-specific accuracies, AAs, OAs, and kappa coefficients of comparison methods, on three data sets are given in Tables IX–XI.

The OA curves of different methods in Fig. 9 show that compared with other methods, the proposed DSRG-CP can obtain competitive results. Specifically, the average OAs

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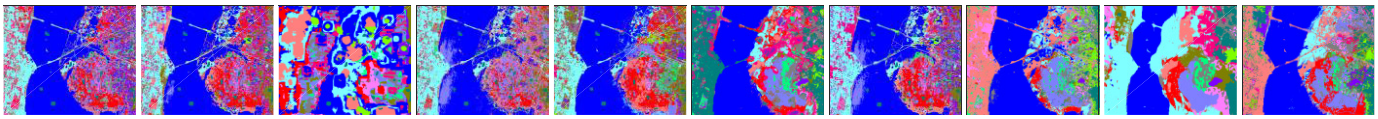


Fig. 10. Classification maps of the comparison methods on KSC images when five samples per class are labeled. (a) CNN (OA = 70.43%). (b) SAE (OA = 73.35%). (c) GFDN (OA = 75.50%). (d) MSRC (OA = 83.03%). (e) SS-RN (OA = 87.25%). (f) HiFi-We (OA = 88.91%). (g) D-SCNN-ATM (OA = 90.29%). (h) LPE (OA = 92.35%). (i) PCA-EPFs (OA = 93.26%). (j) DSRG-CP (OA = 93.74%).

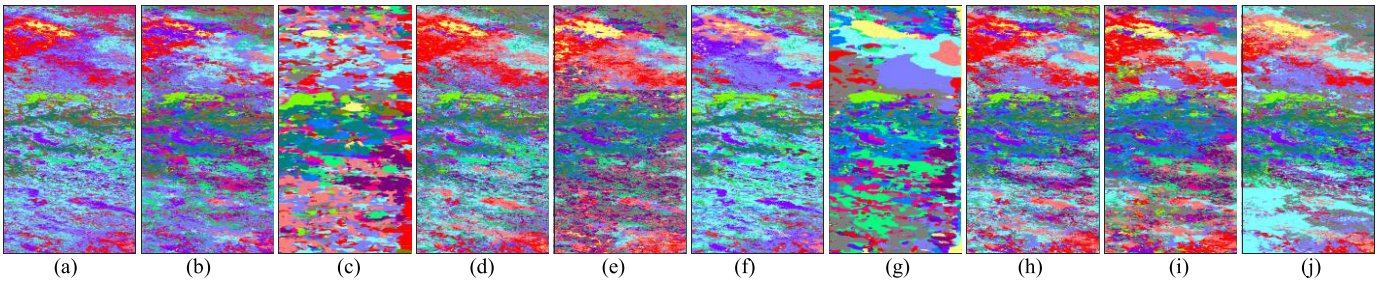


Fig. 11. Classification maps of comparison methods on BOT images when five samples per class are labeled. (a) CNN (OA = 76.66%). (b) SAE (OA = 78.35%). (c) GFDN (OA = 83.19%). (d) MSRC (OA = 86.61%). (e) SS-RN (OA = 89.34%). (f) HiFi-We (OA = 89.49%). (g) PCA-EPFs (OA = 90.65%). (h) D-SCNN-ATM (OA = 90.89%). (i) LPE (OA = 91.25%). (j) DSRG-CP (OA = 94.36%).

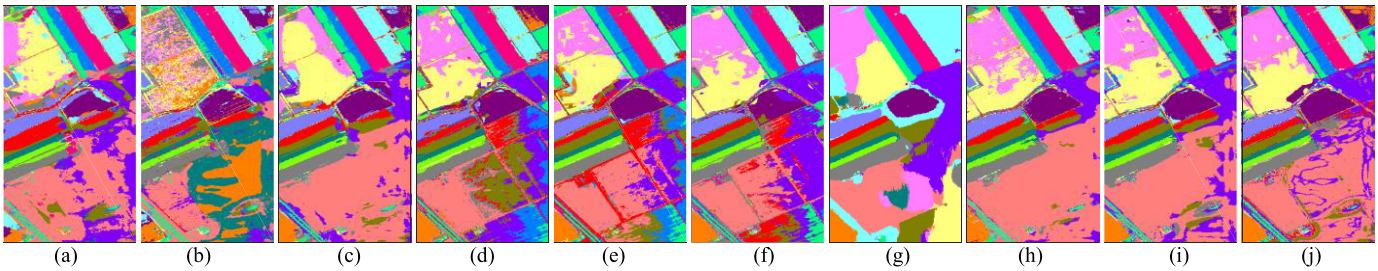


Fig. 12. Classification maps of comparison methods on SV images when five samples per class are labeled. (a) CNN (OA = 66.03%). (b) SAE (OA = %). (c) GFDN (OA = 78.53%). (d) HiFi-We (OA = 85.39%). (e) MSRC (OA = 86.65%). (f) SS-RN (OA = 90.12%). (g) PCA-EPFs (OA = 90.95%). (h) LPE (OA = 91.15%). (i) D-SCNN-ATM (OA = 91.45%). (j) DSRG-CP (OA = 92.46%).

TABLE IX

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE KSC IMAGES WHEN FIVE SAMPLES PER CLASS ARE LABELED

Class No	CNN	SAE	GFDN	MSRC	SS-RN	HiFi-We	LPE	PCA-EPFs	D-SCNN-ATM	DSRG-CP
1	55.56 ± 5.21	85.49 ± 2.32	82.30 ± 2.25	76.62 ± 2.31	87.63 ± 3.87	95.50 ± 2.96	93.83 ± 3.24	97.27 ± 2.21	96.65 ± 1.78	83.78 ± 3.21
2	43.28 ± 3.25	67.23 ± 6.62	84.26 ± 3.17	50.64 ±10.65	98.14 ± 1.06	89.08 ± 3.15	92.11 ± 2.86	99.59 ± 0.35	98.23 ± 1.24	90.58 ± 1.85
3	70.92 ± 4.14	97.58 ± 3.81	77.02 ± 4.86	85.08 ± 2.84	78.23 ± 4.51	96.57 ± 2.40	95.85 ± 2.77	99.49 ± 0.27	95.97 ± 2.26	98.73 ± 1.26
4	91.90 ± 2.55	65.59 ± 7.47	64.92 ± 5.25	84.68 ± 3.65	76.92 ± 6.84	57.09 ± 7.61	84.81 ± 3.15	89.83 ± 2.14	97.52 ± 1.74	99.54 ± 0.20
5	99.21 ± 0.79	79.74 ± 2.84	81.65 ± 3.68	91.77 ± 2.14	100 ± 0.00	95.17 ± 2.84	98.63 ± 1.66	91.41 ± 2.66	95.15 ± 1.35	98.71 ± 0.89
6	74.55 ± 3.25	67.86 ± 4.76	98.52 ± 1.73	88.39 ± 3.35	83.04 ± 4.25	89.73 ± 2.94	93.93 ± 2.91	77.70 ± 3.98	97.05 ± 1.44	93.75 ± 1.32
7	100 ± 0.00	100 ± 0.00	100 ± 0.00	99.36 ± 0.71	100 ± 0.00	100 ± 0.00	100 ± 0.00	91.29 ± 2.21	100 ± 0.00	100 ± 0.00
8	68.08 ± 5.15	43.66 ± 11.02	79.71 ± 7.65	51.29 ± 9.90	92.38 ± 1.75	72.39 ± 6.87	86.75 ± 2.45	90.64 ± 1.33	98.92 ± 1.19	89.78± 1.54
9	61.75 ± 6.21	74.76 ± 3.78	50.39 ± 8.62	82.36 ± 3.87	80.74 ± 1.87	83.46 ± 4.22	93.17 ± 2.69	99.52 ± 0.30	83.31± 3.65	97.60 ± 1.36
10	63.91 ± 3.52	59.15 ± 7.09	73.93 ± 4.32	96.50 ± 2.14	98.50 ± 1.14	87.22 ± 2.93	98.14 ± 0.97	87.58 ± 2.96	99.75 ± 0.23	100 ± 0.00
11	93.72 ± 5.14	44.53 ± 5.45	73.56 ± 512	98.52 ± 1.73	97.34 ± 2.01	91.06 ± 1.96	90.35 ± 3.14	90.71 ± 3.51	92.95 ± 2.48	99.21 ± 0.36
12	51.81 ± 7.25	82.73 ± 3.77	55.31 ± 4.29	92. 57± 2.51	83.90 ± 2.55	91.17 ± 1.85	77.36 ± 6.31	96.23 ± 3.21	95.65 ± 1.78	97.71 ± 1.17
13	84.27 ± 5.88	91.84 ± 2.51	86.94 ± 2.60	85.69 ± 3.35	80.30 ± 2.86	99.35 ± 0.45	99.81 ± 0.98	100 ± 0.00	83.35 ± 3.69	96.89 ± 2.13
AA	73.83 ± 3.21	74.09 ± 2.51	77.50 ± 3.01	83.27 ±3.31	89.09 ± 2.14	88.35 ± 2.17	92.57 ±2.14	93.18 ±1.89	93.11 ± 1.77	95.54 ± 2.34
OA	70.48 ± 2.71	73.37± 2.36	75.58 ± 2.94	82.95 ± 3.08	87.23 ± 2.51	88.91 ± 2.25	92.36 ± 1.87	93.24 ± 2.02	90.32 ± 1.54	93.75 ± 1.81
kappa	67.55 ± 2.96	70.52 ± 2.54	72.63 ± 2.82	80.48 ± 2.95	85.89 ± 2.26	88.34 ± 1.84	91.64 ± 1.64	92.51 ± 1.75	89.82 ± 1.62	93.78 ± 1.56

of DSRG-CP are higher than that of other methods on the KSC and BOT data sets, only slightly lower than that of PCA-EPFs on SV data set. PCA-EPFs obtain good performance lies in the strong edge-preserving filter of PCA-EPFs, which can remove noise, weak edges, and small details in the HSI. For the SV images, the distribution of ground objects is continuous and uniform, and there is a spatial interval between different ground objects, which

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TABLE X
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE BOT IMAGES WHEN FIVE SAMPLES PER CLASS ARE LABELED

Class No	CNN	SAE	GFDN	MSRC	SS-RN	HiFi-We	LPE	PCA-EPFs	D-SCNN-ATM	DSRG-CP
1	97.71 ± 2.35	98.25 ± 1.75	69.43 ± 2.41	98.71 ± 1.35	87.02 ± 3.14	98.75 ± 2.36	86.92 ± 3.51	91.00 ± 2.28	99.02 ± 0.35	98.06 ± 1.24
2	91.40 ± 1.78	98.84 ± 1.25	81.25 ± 2.66	97.18 ± 1.87	99.14 ± 0.26	97.18 ± 2.65	92.31 ± 2.36	93.89 ± 3.01	98.97 ± 0.96	99.25 ± 0.81
3	79.42 ± 2.54	86.61 ± 3.54	86.40 ± 3.07	86.43± 3.26	87.65 ± 1.78	90.61 ± 3.65	79.84 ± 3.54	94.97 ± 2.50	77.87 ± 3.68	97.14 ± 2.21
4	78.74 ± 3.50	87.68 ± 2.58	92.33 ± 1.05	93.51 ± 2.66	98.58 ± 1.05	96.77± 2.74	98.96 ± 1.84	83.94 ± 3.41	98.76 ± 1.56	98.25 ± 0.00
5	76.25 ± 2.94	63.16 ± 4.59	65.53 ± 6.94	75.73 ± 2.69	71.54 ± 5.41	77.25 ± 3.96	57.53 ± 5.76	87.09 ± 1.79	81.69 ± 2.96	95.26 ± 2.14
6	62.45 ± 3.74	70.77 ± 7.74	89.39 ± 4.22	66.11 ±7.81	89.27 ± 4.27	83.43± 4.87	86.10 ± 2.36	82.01 ± 2.64	63.86 ± 5.81	95.98 ± 2.07
7	85.84 ± 2.16	96.81 ± 1.80	92.91 ± 2.08	95.43 ± 2.12	99.61 ± 0.66	92.69 ± 3.38	99.06 ± 0.65	99.67 ± 0.23	97.99 ± 1.85	99.47 ± 0.96
8	87.26 ± 2.28	67.53 ± 2.84	90.40 ± 2.99	95.38 ± 1.74	72.82 ± 5.95	92.31 ± 4.26	97.93 ± 2.74	95.44 ± 1.25	92.23 ± 2.54	98.24 ± 1.79
9	67.97 ± 3.78	71.24 ± 3.21	74.11 ± 4.62	77.78 ± 3.21	87.17 ± 5.12	91.27 ± 5.14	97.25 ± 1.52	92.33 ± 2.24	85.59 ± 3.16	97.77 ± 1.58
10	78.33 ± 4.26	65.27 ± 5.96	99.18 ± 0.58	73.39 ± 4.51	99.14 ± 0.88	84.33 ± 4.95	97.48 ± 1.87	86.19± 2.32	99.02 ± 0.36	99.95± 0.32
11	65.22 ± 5.87	75.08 ± 5.04	99.33 ± 0.66	89.45 ± 2.95	100 ± 0.00	89.81 ± 5.21	100 ± 0.00	100 ± 0.00	100 ± 0.00	98.60 ± 1.34
12	97.13 ± 2.04	97.63 ± 2.11	86.52 ± 1.45	92.72 ± 1.96	98.75 ± 0.96	90.01 ± 3.36	92.23 ± 2.66	96.28 ± 1.87	97.08 ± 1.84	96.61 ± 2.68
13	47.22 ± 9.84	55.60 ± 8.83	79.47 ± 3.97	85.41 ± 2.77	82.08 ± 2.69	91.80 ± 2.36	99.01 ± 0.87	96.23 ± 1.96	99.21 ± 0.71	100 ± 0.00
14	80.46 ± 3.21	97.67 ± 2.04	76.67 ± 4.26	96.92 ± 1.45	95.35 ± 2.71	92.73 ± 2.95	88.24 ± 3.21	96.90 ± 2.34	93.94 ± 2.63	98.47 ± 1.93
AA	77.98 ± 2.84	81.00 ± 2.95	84.17 ± 2.31	87.53 ± 2.15	90.10 ± 1.95	90.17 ± 2.04	91.25 ± 2.34	92.57 ± 1.87	91.70 ± 1.89	94.87 ± 1.76
OA	76.31 ± 2.65	78.41 ± 2.84	83.29 ± 2.25	86.25 ± 1.96	89.05 ± 1.89	89.47 ± 1.98	91.26 ± 2.14	90.67 ± 1.69	90.89 ± 1.74	94.34 ± 1.84
kappa	73.92 ± 2.71	76.68 ± 2.79	83.29 ± 2.40	84.88 ± 2.03	89.19 ± 1.77	88.92 ± 2.13	90.42 ± 1.89	89.89 ± 1.71	90.15 ± 1.75	94.25 ± 1.62

TABLE XI
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE SV IMAGES WHEN FIVE SAMPLES PER CLASS ARE LABELED

Class No	CNN	SAE	GFDN	MSRC	SS-RN	HiFi-We	LPE	PCA-EPFs	D-SCNN-ATM	DSRG-CP
1	98.25 ± 1.36	95.51 ± 1.36	99.03 ± 1.14	99.81 ± 2.14	92.93 ± 2.54	94.19 ± 3.25	99.75 ± 2.55	100 ± 0.00	100 ± 0.00	98.62 ± 1.12
2	56.31 ± 3.58	86.81 ± 1.78	88.81 ± 0.74	95.02 ± 1.78	76.20 ± 1.87	90.17 ± 2.14	97.29 ± 3.16	99.25 ± 0.32	93.93 ± 3.32	98.59 ± 0.97
3	57.85 ± 3.79	57.27 ± 2.21	74.40 ± 3.25	99.13 ± 0.25	98.46 ± 2.55	96.28 ± 3.41	99.02 ± 0.54	84.87 ± 2.65	99.64 ± 0.56	97.63 ± 1.55
4	79.15 ± 5.81	58.76 ± 3.67	99.21 ± 3.21	98.99 ± 1.14	93.38 ± 3.14	93.34 ± 5.22	99.85 ± 0.11	84.74 ± 3.90	99.27 ± 0.37	99.38 ± 0.61
5	98.70 ± 1.85	92.71 ± 4.28	80.49 ± 2.41	97.15 ± 2.68	96.60 ± 2.70	93.32 ± 2.48	95.52 ± 1.01	99.86 ±0.10	97.71 ± 1.15	97.02 ± 3.21
6	97.22 ± 3.12	98.69 ± 2.45	97.75 ± 4.82	97.95 ± 3.21	92.87 ± 3.39	91.69 ± 3.17	99.63 ± 0.29	98.17 ± 0.43	99.67 ± 0.33	98.49 ± 1.35
7	98.96 ± 2.25	97.56 ± 3.65	85.81 ± 2.68	94.44 ± 2.32	93.10 ± 1.85	96.97 ± 1.95	98.25 ± 0.57	96.21 ± 1.25	97.72 ± 1.47	99.43 ± 0.16
8	60.65 ± 7.88	34.83 ± 2.98	35.03 ± 10.51	66.37 ± 9.87	79.05 ± 2.36	65.70 ± 7.55	71.03 ± 7.34	93.61 ± 2.11	80.54 ± 1.84	81.43 ± 3.21
9	58.69 ± 2.63	83.95 ± 3.20	98.95 ± 1.12	97.16 ± 1.02	98.75 ± 3.47	97.40 ± 3.17	98.76 ± 2.14	98.06 ± 1.02	97.80 ± 2.14	99.77 ± 0.21
10	78.78 ± 5.74	85.28 ± 4.74	86.79 ± 5.32	97.37 ± 2.25	87.07 ± 3.47	87.08 ± 4.35	92.45 ± 4..21	95.70 ± 2.33	97.67 ± 1.87	96.84 ± 3.04
11	81.99 ± 6.12	88.54 ± 2.36	99.62 ± 5.65	97.15 ± 1.73	99.75 ± 0.44	99.52 ± 0.36	95.16 ± 2.66	90.07 ± 3.21	99.33 ± 0.96	99.71 ± 0.25
12	57.97 ± 2.95	70.43 ± 3.35	89.58 ± 1.74	96.03 ± 2.34	94.66 ± 3.27	89.19 ± 2.65	98.43 ± 1.04	94.74 ± 2.54	93.97 ± 3.22	97.84 ± 1.35
13	80.53 ± 4.78	70.87 ± 2.07	83.39 ± 3.86	98.23 ± 2.08	95.54 ± 5.81	99.67 ± 0.58	99.55 ± 0.17	88.45 ± 3.09	99.65 ± 0.54	100 ± 0.00
14	84.43 ± 2.01	95.25 ±3.22	98.15 ± 1.07	99.71 ± 1.33	99.95 ± 0.69	99.52 ± 0.71	88.99 ± 2.74	93.75 ± 2.65	96.25 ± 1.77	99.41 ± 0.54
15	27.54 ± 10.25	35.98 ± 9.65	83.67 ± 7.35	68.78 ± 3.78	90.99 ± 2.08	73.88 ± 3.54	85.07 ± 5.21	74.24 ± 6.32	79.73 ± 4.48	94.88 ± 3.12
16	63.10 ± 3.37	64.87 ± 4.31	84.11 ± 5.41	97.66 ± 3.11	98.31 ± 1.15	94.87 ± 2.04	93.74 ± 3.01	100 ±0.00	99.74 ± 0.37	97.87 ± 0.95
AA	73.83 ± 2.58	76.03 ± 2.54	86.67 ± 1.84	93.91 ± 1.78	92.97 ±1.45	91.46 ± 1.74	94.55 ± 2.14	93.23 ± 1.61	95.78 ± 2.01	96.92 ± 1.72
OA	65.93 ± 2.39	68.02 ± 2.28	78.38 ± 2.16	86.63 ± 2.35	90.06 ± 1.93	85.32 ± 2.12	91.02 ± 1.96	90.57 ± 1.28	91.42 ± 2.14	92.51 ± 1.56
kappa	62.89 ± 2.25	64.79 ± 2.03	75.85 ± 1.93	84.13 ± 1.55	88.79 ± 2.17	83.88 ± 1.82	89.59 ± 1.46	89.44 ± 1.17	90.87 ± 1.83	92.06 ± 1.34

makes the classification results on the reference map more accurate.

An interesting result is that deep learning-based approaches are more sensitive to the number of labeled samples, when the number of labeled samples is limited, and the average OAs are very low. Also, for deep learning-based methods, GFDN achieves better classification performance than CNN and SAE since GFDN uses the spatial features to train the model. MSRC, SS-RN, and D-SCNN-ATM are specifically designed for limited training samples. These methods obtain better classification performance than that of CNN and SAE, but the classification performance is lower than that of DSRG-CP. The good performance of DSRG-CP is derived from two aspects. One is the pseudolabel mechanism of copropagation. The other is the effectiveness of spectral and spatial SR graphs.

The results from Figs. 10–12 demonstrate that DSRG-CP performs better performance visually even only five labeled samples are given. The results in Tables IX–XI show that DSRG-CP obtains much higher accuracies in terms of AAs, OAs, and kappa coefficients than CNN, SAE, GFDN, MSRC, SS-RN, and HiFi-We. The classification accuracy of DSRG-CP is a little higher than that of LPE, PCA-EPFs, and D-SCNN-ATM. This is because these methods adopt more excellent techniques to extract features or use more sophisticated models. For examples, LPE combines extended multiattribute profiles, semisupervised method, and ensemble method together to build models. PCA-EPFs adopt strong edge-preserving filter, which can remove noise, weak edges, and small details in the HIS. D-SCNN-ATM is a combination framework of extended morphological profiles, CNN, siamese network, and spectral-spatial feature fusion, which integrates several techniques to increase the classification performance.

F. Parameter Sensitivity Analysis

In previous experiments, we set the parameters α and N to 0.4 and 1500, respectively. The parameter α is used to balance the significance between the SPE-G and the SPA-G,

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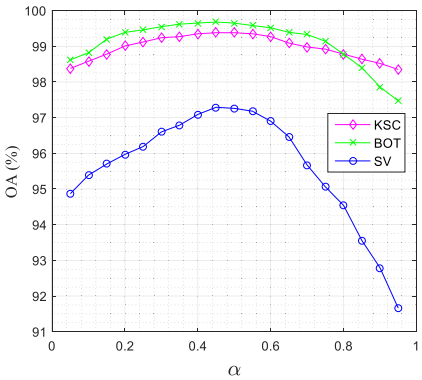


Fig. 13. Influence of the parameter α on the classification performance.

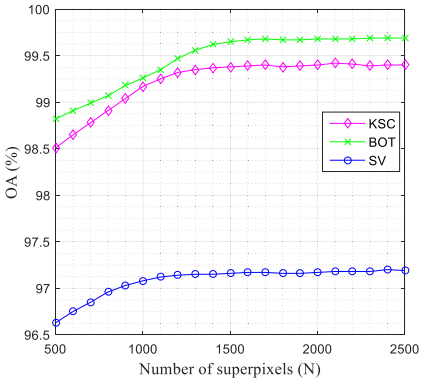


Fig. 14. Influence of the parameter N on the classification performance.

and the parameter N represents the number of superpixels. In this section, we analyze the parameters α and N. When we analyze one parameter, the other parameter is fixed.

The influence of the parameter α on the classification performance is shown in Fig. 13. The results in Fig. 13 indicate that the parameter α is robust to the classification performance, and α has a wide range of values. The suggested value of the parameter α belongs to [0.3, 0.6]. The reason for this result is that although both the SPE-G and the SPA-G are not accurate enough, the copropagation mechanism obtains more labeled samples for label propagation, which finally makes the results of decision fusion more accurate.

Fig. 14 shows the influence of parameter N on the classification accuracy. The OA curves in Fig. 14 exhibit that the OA increases with the increase of N and then tends to be stable. In other words, when the number of superpixels is too small, each superpixel block has too many pixels, which may make the superpixel block be mixed. We aim to use the superpixel to represent one pixel in the superpixel, so the superpixel with a small number of pixels will be helpful.

G. Further Analysis of DSRG-CP

In this section, we do more experiments to analyze the proposed method. To demonstrate the significance of copropagation, we use self-training for the iteration of label propagation. In detail, we fuse the spectral graph $\mathbf{W}^1$ and the spatial graph $\mathbf{W}^2$ into spectral-spatial graph. Then, we perform label propagation on the fused graph iteratively. We call this method self-training-based label propagation (STLP). The adjacency matrix of the fused graph can be written as follows:

$$\mathbf{W} = \mu\mathbf{W}^1 + (1-\mu)\mathbf{W}^2. \tag{35}$$

According to [21], we set parameter μ to 0.2 for experiments.

Recently, SuperPCA [60] was proposed for unsupervised feature extraction, which uses the superpixelwise technique to avoid destroying the discriminant of data. To further explore the performance of DSRG-CP under the situation of discriminative dimensionality reduction, we use SuperPCA instead of PCA for feature extraction before extracting 2-D Gabor features.

TABLE XII
OVERALL ACCURACIES OF THREE METHODS WHEN FIVE SAMPLES PER CLASS ARE LABELED

Data set	STLP	DSRGCP (PCA)	DSRG-CP(SuperPCA)
KSC	91.78 ± 1.95	93.75 ± 1.81	94.88 ± 1.74
BOT	91.45 ± 1.79	94.34 ± 1.84	95.73 ± 1.86
SV	90.85 ± 1.76	92.51 ± 1.56	94.12 ± 1.73

To show the obvious differences between the different approaches, we randomly select five labeled samples per class to form the training set. We do experiments ten times for each method, and the training samples are randomly selected at each time. The average OAs of ten runs of experiments can be found in Table XII. Compared with STLP, DSRG-CP increases OAs obviously, which demonstrates that cotraining used for the iteration of label propagation is more useful than that of self-training. In addition, SuperPCA instead of PCA used in DSRG-CP can further increase the classification performance. In other words, a good feature extraction method is helpful to the classification, although it is only used for preprocessing.

V. CONCLUSION

In this article, a DSRR-CP method is proposed for semisupervised HSI classification. The experiments on several HSI data sets demonstrate that DSRG-CP can achieve a competitive solution for semisupervised HSI classification, especially in the case of very limited labeled samples. We conclude several points from previous experimental results. First, SR graphs are obviously better than traditional graphs. Second, the regularizers in both spectral graph and spatial graph play significant roles in increasing the classification accuracy. Third, the copropagation is very effective. It makes the final probability matrices of both spectral graph and spatial graph more accurate, which increases the accuracy of decision fusion. At last, DSRG-CP is robust to parameters, and a wide range of values can meet the requirements. In addition, the classification performance of DSRG-CP can be further improved by using more advanced preprocessing techniques such as SuperPCA.

Meanwhile, the limitations of the proposed method are summarized as follows. First, the proposed approach requires more computational cost than other graph-based methods. Second, the copropagation strategy decreases the class-specific

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accuracy on some classes. At last, the proposed method needs to fine-tune several parameters.

In future work, we consider adopting the anchor graph to speed up DSRG-CP. Also, constructing other graphs such as low-rank and collaborative graphs is considerable.

REFERENCES

[1] L. Ma, M. M. Crawford, and J. Tian, "Local manifold learning-based k-nearest-neighbor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4099–4109, Nov. 2010.

[2] B. Tu, S. Huang, L. Fang, G. Zhang, J. Wang, and B. Zheng, "Hyperspectral image classification via weighted joint nearest neighbor and sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4063–4075, Nov. 2018.

[3] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[4] Y. Tarabalka, M. Fauvel, J. Chanussot, and J. A. Benediktsson, "SVM- and MRF-based method for accurate classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 4, pp. 736–740, Oct. 2010.

[5] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification via kernel sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 217–231, Jan. 2013.

[6] W. Li, C. Chen, H. Su, and Q. Du, "Local binary patterns and extreme learning machine for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3681–3693, Jul. 2015.

[7] Y. Bazi, N. Alajlan, F. Melgani, H. AlHichri, S. Malek, and R. R. Yager, "Differential evolution extreme learning machine for the classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 6, pp. 1066–1070, Jun. 2014.

[8] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[9] Y. Zhang, G. Cao, X. Li, and B. Wang, "Cascaded random forest for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1082–1094, Apr. 2018.

[10] G. Hughes, "On the mean accuracy of statistical pattern recognizers," *IEEE Trans. Inf. Theory*, vol. IT-14, no. 1, pp. 55–63, Jan. 1968.

[11] X. Zhu, "Semi-supervised learning literature survey," Univ. Wisconsin-Madison, Madison, WI, USA, Tech. Rep. 1530, Jul. 2006.

[12] I. Dópido, J. Li, P. R. Marpu, A. Plaza, J. M. B. Dias, and J. A. Benediktsson, "Semisupervised self-learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 7, pp. 4032–4044, Jul. 2013.

[13] F. Li, D. A. Clausi, L. Xu, and A. Wong, "ST-IRGS: A region-based self-training algorithm applied to hyperspectral image classification and segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 3–16, Jan. 2018.

[14] Q. Jackson and D. A. Landgrebe, "An adaptive classifier design for high-dimensional data analysis with a limited training data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 12, pp. 2664–2679, Dec. 2001.

[15] L. Bruzzone, M. Chi, and M. Marconcini, "A novel transductive SVM for semisupervised classification of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 11, pp. 3363–3373, Nov. 2006.

[16] G. Camps-Valls, T. V. B. Marsheva, and D. Zhou, "Semi-supervised graph-based hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3044–3054, Oct. 2007.

[17] Y. Gao, R. Ji, P. Cui, Q. Dai, and G. Hua, "Hyperspectral image classification through bilayer graph-based learning," *IEEE Trans. Image Process.*, vol. 23, no. 7, pp. 2769–2778, Jul. 2014.

[18] W. Wang and Z. H. Zhou, "A new analysis of co-training," in *Proc. 27th Int. Conf. Mach. Learn.*, Haifa, Israel, 2010, pp. 1135–1142.

[19] M. Romaszewski, P. Głomb, and M. Cholewa, "Semi-supervised hyperspectral classification from a small number of training samples using a co-training approach," *ISPRS J. Photogramm. Remote Sens.*, vol. 121, pp. 60–76, Nov. 2016.

[20] S. S. Sawant and M. Prabukumar, "A review on graph-based semi-supervised learning methods for hyperspectral image classification," *Egyptian J. Remote Sens. Space Sci.*, vol. 23, no. 2, pp. 243–248, Aug. 2020.

[21] L. Wang, S. Hao, Q. Wang, and Y. Wang, "Semi-supervised classification for hyperspectral imagery based on spatial–spectral label propagation," *ISPRS J. Photogramm. Remote Sens.*, vol. 97, pp. 123–137, Nov. 2014.

[22] L. Ma, M. M. Crawford, X. Yang, and Y. Guo, "Local-manifold-learning-based graph construction for semisupervised hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2832–2844, May 2015.

[23] Y. Shao, N. Sang, C. Gao, and L. Ma, "Probabilistic class structure regularized sparse representation graph for semi-supervised hyperspectral image classification," *Pattern Recognit.*, vol. 63, pp. 102–114, Mar. 2017.

[24] L. Ma, A. Ma, C. Ju, and X. Li, "Graph-based semi-supervised learning for spectral–spatial hyperspectral image classification," *Pattern Recognit. Lett.*, vol. 83, pp. 133–142, Nov. 2016.

[25] N. Jamshidpour, A. Safari, and S. Homayouni, "Spectral–spatial semi-supervised hyperspectral classification using adaptive neighborhood," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 9, pp. 4183–4197, Sep. 2017.

[26] Y. Shao, N. Sang, C. Gao, and L. Ma, "Spatial and class structure regularized sparse representation graph for semi-supervised hyperspectral image classification," *Pattern Recognit.*, vol. 81, pp. 81–94, Sep. 2018.

[27] H. Li, Y. Wang, S. Xiang, J. Duan, F. Zhu, and C. Pan, "A label propagation method using spatial–spectral consistency for hyperspectral image classification," *Int. J. Remote Sens.*, vol. 37, no. 1, pp. 191–211, Jan. 2016.

[28] S. Jia, X. Zhang, and Q. Li, "Spectral–spatial hyperspectral image classification using $\ell 1/2$ regularized low-rank representation and sparse representation-based graph cuts," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2473–2484, Jun. 2015.

[29] B. Cui, X. Xie, S. Hao, J. Cui, and Y. Lu, "Semi-supervised classification of hyperspectral images based on extended label propagation and rolling guidance filtering," *Remote Sens.*, vol. 10, no. 4, p. 515, Mar. 2018.

[30] X. Zhu, Z. Ghahramani, and J. D. Lafferty, "Semi-supervised learning using Gaussian fields and harmonic functions," in *Proc. 20th Int. Conf. Mach. Learn.*, Washington, DC, USA, 2003, pp. 912–919.

[31] D. Zhou, O. Bousquet, T. Lal, J. Weston, and B. Schölkopf, "Learning with local and global consistency," in *Proc. Adv. Neural Inf. Process. Syst.*, Vancouver, BC, Canada, 2004, pp. 321–328.

[32] F. Wang and C. Zhang, "Label propagation through linear neighborhoods," *IEEE Trans. Knowl. Data Eng.*, vol. 20, no. 1, pp. 55–67, Jan. 2008.

[33] Y. Zhang, G. Cao, A. Shafique, and P. Fu, "Label propagation ensemble for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3623–3636, Sep. 2019.

[34] H. Cheng, Z. Liu, and J. Yang, "Sparsity induced similarity measure for label propagation," in *Proc. IEEE 12th Int. Conf. Comput. Vis.*, Kyoto, Japan, Sep. 2009, pp. 317–324.

[35] S. Yan and H. Wang, "Semi-supervised learning by sparse representation," in *Proc. Int. Conf. Data Mining (SIAM)*, Sparks, NV, USA, Apr. 2009, pp. 788–797.

[36] Y. Gu and K. Feng, "L1-graph semisupervised learning for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Munich, Germany, Jul. 2012, pp. 1401–1404.

[37] N. H. Ly, Q. Du, and J. E. Fowler, "Sparse graph-based discriminant analysis for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 3872–3884, Jul. 2014.

[38] K. Tan, S. Zhou, and Q. Du, "Semisupervised discriminant analysis for hyperspectral imagery with block-sparse graph," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 8, pp. 1765–1769, Aug. 2015.

[39] J. Bai, S. Xiang, and C. Pan, "A graph-based classification method for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 2, pp. 803–817, Feb. 2013.

[40] A. Appice, P. Guccione, and D. Malerba, "A novel spectral–spatial co-training algorithm for the transductive classification of hyperspectral imagery data," *Pattern Recognit.*, vol. 63, pp. 229–245, Mar. 2017.

[41] X. Zhang, Q. Song, R. Liu, W. Wang, and L. Jiao, "Modified co-training with spectral and spatial views for semisupervised hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2044–2055, Jun. 2014.

[42] S. Samiappan and R. J. Moorhead, "Semi-supervised co-training and active learning framework for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Milan, Italy, Jul. 2015, pp. 401–404.

[43] A. Blum and T. Mitchell, "Combining labeled and unlabeled data with co-training," in *Proc. 11th Annu. Conf. Comput. Learn. Theory*, Madison, WI, USA, 1998, pp. 92–100.

[44] C.-I. Chang, "An information-theoretic approach to spectral variability, similarity, and discrimination for hyperspectral image analysis," *IEEE Trans. Inf. Theory*, vol. 46, no. 5, pp. 1927–1932, Aug. 2000.

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[45] Z. Lin, R. Liu, and Z. Su, "Linearized alternating direction method with adaptive penalty for low-rank representation," in *Proc. Adv. Neural Inf. Process. Syst.*, Granada, Spain, 2011, pp. 612–620.
[46] M. Belkin, P. Niyogi, and V. Sindhwani, "Manifold regularization: A geometric framework for learning from labeled and unlabeled examples," *J. Mach. Learn. Res.*, vol. 7, pp. 2399–2434, Nov. 2006.
[47] Y. X. Wang and H. Xu, "Noisy sparse subspace clustering," in *Proc. 30th Int. Conf. Mach. Learn.*, Atlanta, GA, USA, 2013, pp. 89–97.
[48] F. van der Meer, "The effectiveness of spectral similarity measures for the analysis of hyperspectral imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 8, no. 1, pp. 3–17, Jan. 2006.
[49] T. P. Weldon, W. E. Higgins, and D. F. Dunn, "Efficient Gabor filter design for texture segmentation," *Pattern Recognit.*, vol. 29, no. 12, pp. 2005–2015, Dec. 1996.
[50] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "Slic superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2281, Nov. 2012.
[51] T. Eiter and H. Mannila, "Distance measures for point sets and their computation," *Acta Inf.*, vol. 34, no. 2, pp. 109–133, Feb. 1997.
[52] X. Kang, C. Li, S. Li, and H. Lin, "Classification of hyperspectral images by Gabor filtering based deep network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1166–1178, Apr. 2018.
[53] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, Jul. 2015, Art. no. 258619.
[54] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.
[55] B. Pan, Z. Shi, and X. Xu, "Hierarchical guidance filtering-based ensemble classification for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 4177–4189, Jul. 2017.
[56] X. Kang, X. Xiang, S. Li, and J. A. Benediktsson, "PCA-based edge-preserving features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7140–7151, Dec. 2017.
[57] M. Rao, P. Tang, and Z. Zhang, "Spatial–spectral relation network for hyperspectral image classification with limited training samples," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 5086–5100, Dec. 2019.
[58] B. Pan, Z. Shi, and X. Xu, "Multiobjective-based sparse representation classifier for hyperspectral imagery using limited samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 239–249, Jan. 2019.
[59] L. Huang and Y. Chen, "Dual-path siamese CNN for hyperspectral image classification with limited training samples," *IEEE Geosci. Remote Sens. Lett.*, early access, Mar. 19, 2020, doi: 10.1109/LGRS.2020.2979604.
[60] J. Jiang, J. Ma, C. Chen, Z. Wang, Z. Cai, and L. Wang, "SuperPCA: A superpixelwise PCA approach for unsupervised feature extraction of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4581–4593, Aug. 2018.



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